Re-Implemented, Not Transported: Auditing Cross-Paper Comparisons in Robust RAG

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Abstract

Robust retrieval-augmented generation (RAG) papers share labels—noise robustness, faithfulness, security, calibration—while answering different decision questions, so a quoted cross-paper gap may not be a comparison at all. We ask what an automated audit of such gaps can and cannot decide, and we answer it by varying the evidence rather than only the model. A documented screen over 259 full texts yields 1,703 candidate juxtapositions, coded independently by judges from three developers. One judgment is recovered: whether a number was *transported* from the other paper or *re-implemented* in this harness. Judges agree on it (κ 0.55) and, scored on 180 *real* windows against a reference set by adjudicators that read both papers' full texts, the deployed panel reaches 0.91 [0.86, 0.96] against a 0.79 baseline. In this frame the distinction is lopsided: 533 of the 610 candidates assigned a class re-implement the other system and 77 transport a number. The adjudication is not recovered: verdict κ reaches 0.31 for an all-frontier panel drawn from a fully crossed pool of 9, composing the decision rule from the judges' own coordinate labels does not help (+0.02 [0.00, 0.05]), and neither does simplifying the schema. Three interventions then locate the cause instead of assuming it. Giving the same judges the entire citing paper instead of a 45-line window raises verdict agreement by +0.20 [0.09, 0.36], so a large part of the census figure is an artefact of what a screen can see—but it lands at *fair*, not at reproducible. Giving them an explicit *the passage does not say* option, which they use on 42.4% of codings, moves it by +0.01 [-0.07, 0.11], so the instrument was not the obstacle. What closes the gap is neither: on constructed passages that state their coordinates the same instrument reaches κ 0.74, at 92% of the ceiling its own marginals allow against 13% on published text. Disclosure helps a reader who has the papers and does not make the comparison reproducible, and the two facts together are what the coordinates should be read as recommending. We release the documented frame, all codings, four verifiers and every simulation.

1 Introduction

Retrieval-augmented generation conditions a generator on retrieved evidence (Lewis et al., 2020; Karpukhin et al., 2020; Guu et al., 2020), and robust-RAG studies ask what happens when that evidence is irrelevant, insufficient, conflicting, stale, weakly sourced, or malicious. Two reported "robustness" gains may nevertheless concern different retrieval snapshots, attacker capabilities, interventions, outcome units, or deployment actions. Source verification is therefore necessary but insufficient: a cross-paper claim must also establish that both results answer a compatible decision question.

One case, in full. PoisonedRAG reports a "97% ASR by injecting 5 malicious texts for each target question" on Natural Questions, and its Table 4 places that beside four baselines, the weakest—the Corpus Poisoning Attack of Zhong et al. (2023)—at 0.01 (Zou et al., 2025). A reader carries away that corpus poisoning does not work. Corpus poisoning's own paper reports it placing an adversarial passage in the retrieved set for 84.2% of Natural Questions queries with a *single* adversarial passage, and 99.4% with fifty.

The two settings share a corpus and a retriever and nothing else that matters, and we state each fact where it lives. Both use Natural Questions and the pre-trained Contriever; PoisonedRAG declares this as its default, retrieving "5 most similar texts from the knowledge database as the context" (Zou et al., 2025, l. 681–684), while the 84.2 and 99.4 are the Contriever row of the NQ block at $|\mathcal{A}| = 1$ and 50 (Zhong et al., 2023, l. 366–372). What differs is what is counted: corpus poisoning's success is retrieval—"the percentage of queries for which at least one adversarial passage is retrieved

in the top-k results" (Zhong et al., 2023, l. 333–337), at top-20—while PoisonedRAG's outcome is the fraction of target questions answered with the attacker's chosen answer, which corpus poisoning never chooses. PoisonedRAG says so: the three weakest baselines "are ineffective because they cannot achieve generation, generation, and retrieval condition, respectively" (Zou et al., 2025, l. 1338–1341). That is the whole content of the 0.01, and inside PoisonedRAG's harness it is the right number. A released verifier re-opens every line range here and confirms each value and condition (7/7 checks pass).

The two figures are printed in different units, count different events, and range over different retrieved sets, so the outcome coordinate is unmatched and no common scope exists in either text. The comparison is undefined rather than repairable: there is no operating point at which 1% and 84.2% answer one question. What survives is that these two attacks cannot be ranked by these two numbers—and that the paper being benchmarked said as much, in a sentence the number outlived.

The question this paper answers. That case is instructive, but it is one case and it was assembled by hand. Two things are unknown: how often the failure occurs, and whether finding it can be automated so a reviewer could run the check. We build that screen and measure, judgment by judgment, how much of it works. The answer splits sharply. *Reading* a comparison—whether a number was carried across from another paper or produced again locally—is recovered, and validated on real windows against a reference set by adjudicators that read both papers rather than a window. *Adjudicating* it is not recovered, by any panel we can build, any simplification of the instrument, any amount of extra evidence, or an explicit licence to abstain.

The last two are the point. An observational audit cannot distinguish a task that published text underdetermines from judges that cannot report underdetermination, so we stop observing and intervene, holding the judges, the codebook and the items fixed while moving one thing at a time. Give the judges the entire citing paper, then both papers, in place of the window: agreement climbs, by +0.20 [0.09, 0.36], and stops at *fair*. Give them a fourth verdict meaning *the passage does not say*, worded prominently and preferred under uncertainty: they use it freely and agreement does not move. So part of the census failure is what a screen cannot see, none of it is an instrument that cannot express doubt, and the remainder—the distance to the κ 0.74 the same judges reach on passages written to state their coordinates—is what only rewriting the papers would close.

Contributions. (i) A census of cross-paper juxtapositions in robust RAG with the sampling frame documented rather than asserted: a flow from 1,904 retrieved records to 259 extracted full texts to 1,703 candidates (Appendix Figure 1), coded by independent judges from three developers, establishing that 533 of the 610 candidates assigned a provenance class re-implement the other system rather than transporting its number, and that judges agree on this at κ 0.55 while they do not agree on the verdict (§5). (ii) A privileged-information reference on 180 *real* frame windows, adjudicated from both papers' full texts with mechanically located quotes, replacing a constructed validation set on which every judge was perfect and which therefore had no power to detect the bias it was invoked to rule out (§5). (iii) Three interventions that test the paper's causal claim instead of asserting it—an evidence ladder, an explicit-abstention arm and the constructed benchmark—with composed-rule and disclosure analyses that rule out rule execution and bound how much of the failure extra evidence repairs (§5, §6). (iv) A fully crossed tier-by-developer pool of 9 judges analysed at the judge level, a three-arm codebook ablation, a source-verified ledger of twelve worked comparisons, and a between-reviewer heterogeneity estimate over 22 judges—9 of them open-weight checkpoints run in a single harness—reported as an interval rather than a design constant (§4, §7).

Terms used throughout. A *harness* is one paper's corpus snapshot, retriever, generator, metric and evaluation code taken together; a number is *transported* when it crosses a harness boundary and *re-implemented* when the citing paper produced it inside its own. A *window* is a fixed span of lines in one extracted full text, the *frame* is the set of windows a deterministic screen returns, and a *stratum* is a declared subset defined by a document feature or a construction choice, never by an audit outcome. A *coordinate* is one of the five fields of §3. We say *judge* for a model that codes a window or an item and *reviewer* for the abstract unit whose variability §7 is about; every judge here is a reviewer, and no person is. The frame, every judge's raw output,

both constructed benchmarks, three verifiers and every simulation are in the supplementary material.

2 Positioning

Architectures differ in where retrieval enters and how evidence is fused (Chen et al., 2017; Khandelwal et al., 2020; Izacard and Grave, 2021; Borgeaud et al., 2022; Izacard et al., 2023; Shi et al., 2024), and evaluation, trustworthiness, security, conflict and abstention surveys expose different failure surfaces (Gao et al., 2024; Yu et al., 2024; Zhou et al., 2024; Xu et al., 2026, 2024; Wen et al., 2025). Common-harness comparisons improve empirical control (Wang et al., 2024; Alushi et al., 2026) but do not resolve whether a quoted cross-paper result concerns the same evidence state, capability envelope, outcome or action. Our unit is an individual cross-paper juxtaposition (Appendix Table 30). Annotation research separately distinguishes reproducibility from truth (Artstein and Poesio, 2008; Dawid and Skene, 1979), treats disagreement as signal (Uma et al., 2021; Plank, 2022; Davani et al., 2022), and models rater and item effects (Passonneau and Carpenter, 2014; Paun et al., 2018); crossed designs require both sources of variation (Baayen et al., 2008), clustering follows treatment assignment (Abadie et al., 2023), and power belongs to the randomized unit (Card et al., 2020; van Miltenburg et al., 2021). Auditing how evaluations are reported is likewise established (Howcroft et al., 2020; van der Lee et al., 2021; Belz et al., 2021, 2023; Clark et al., 2021), and we follow their reporting conventions, including a PRISMA-style screening flow (Page et al., 2021); we claim no prevalence those studies do not already establish. That inference over the wrong factor inflates type-I error is also not new: Clark (1973) named the language-as-fixed-effect fallacy and Judd et al. (2012) gave the crossed random-effects remedy, and both subsume the general form of our simulation, which we therefore report in Appendix G rather than as a finding. What they do not supply is the parameter, which has never been estimated for comparability adjudication.

3 What a Comparison Has to Fix

Four boundaries motivate the coordinates: evidence availability is not evidence use (Yoran et al., 2024; Fang et al., 2024; Chen et al., 2024a; Yu et al., 2023); support, authority and conflict are distinct, so a response can be correct but unsupported or faithful to a superseded source (Niu et al., 2024; Wang et al., 2025a; Vu et al., 2024); natural retrieval failure and adaptive attack are different games under different access, budget and persistence assumptions (Zou et al., 2025; Chaudhari et al., 2024; Xiang et al., 2024); and a confidence signal is not an action policy (Guo et al., 2017; Geifman and El-Yaniv, 2017; Asai et al., 2024; Yan et al., 2024; Joren et al., 2025). Appendix A develops these with paper-level contrasts. Each side of a comparison is recorded on five coordinates: **C1** *Evidence state*: what is observable at the decision boundary—supported, distracting, insufficient, conflicting, stale, weakly sourced or adversarially manipulated—with the corpus snapshot, retriever, top-k and whether a gold passage is present. **C2** *Origin / capability*: how that state arose—benign retrieval error, a constructed stressor, drift, or an adversary—and for an adversary the access, persistence, and crucially whether the budget counts documents *injected into the corpus* or *retrieved into the context*, and whether the metric runs over all queries or only triggered ones. **C3** *Intervention*: which part of the system changes. **C4** *Outcome and unit*: what is measured and in what unit—answer correctness, retrieval quality, claim support, attack success, leakage, calibration, selective risk or cost, per question, response or span. **C5** *Policy / utility*: which response is selected under which costs—answer, verify, re-retrieve, expose conflict, quarantine or abstain—with the error, abstention and follow-up prices stated.

A claim-specific control check accompanies them (Appendix Table 31); these specialize distinctions from evidence synthesis, selective prediction and security evaluation (Richardson et al., 1995; Chow, 1970; Papernot et al., 2018).

One decision rule, and why C3 is not in it. Record each coordinate on both sides from the sources alone, then apply, in order. **(1)** If C1, C2, C4 and C5 all match, the quoted gap stands (*survives*). **(2)** Otherwise, if a side reports an operating point at the other's setting, or the comparison holds on a strict subset, re-read there and report the matched number (*narrows*). **(3)** Otherwise the gap is *undefined* and must be replaced by the common scope both studies support, or dropped—never by a substitute number computed across an unmatched coordinate.

C3 is deliberately absent from every branch. Call it the *intervention exclusion*: comparing two *differ-*

ent systems means, by construction, that the intervention differs, so C3 is the thing being compared, not a condition on comparability. A rule treating a C3 mismatch as threatening would declare essentially the whole literature undefined, and the data bear this out twice. C3 is labelled "mismatch" on 99.6% of corpus items where the panel reaches a majority—so near-constant that its κ ceiling is 0.42 and the statistic carries no information—and even on constructed passages that state it explicitly, judges agree at κ 0.00 (§6). It is the only coordinate at chance in *both* settings. We record C3 because a reader needs to know what changed, and exclude it from the decision because it always changed.

4 Twelve Worked Comparisons

We assembled comparisons of two kinds from 259 locally extracted full texts: *stated* ones an author made (8), and *survey-invited* juxtapositions a survey's grouping leads a reader to make (4), reported separately. A stated comparison may *transport* the other paper's number, *re-implement* the other system and print its own, or assert a relation while supplying *no* number for the other side. Every value is copied verbatim with a file and line range, and four released checks run over the result: one re-opens each range and fails on any numeral absent from the cited lines; one requires every quoted string this manuscript prints to occur verbatim at a registered range; one checks that every quantity we assert about our own artifacts equals the field that produced it; and one re-verifies the motivating case of §1 condition by condition. All 102 ledger and 7/7 motivating-case checks pass. They establish that our quotations are faithful, not that our *judgments* are right—so 6 judges from 3 developers re-coded every case from the neutral juxtaposition and the excerpts alone, never seeing our labels (Appendix Table 4). The majority reproduces us on 58% of three-way labels (κ 0.32) and on 100% of the binary distinction our conclusions use. We therefore report no rate from the ledger, only what it supports: these comparisons, by name, are not rankings. All five security cases involve an unmatched attacker capability or objective; the audit does not flag everything, and CRAG's 6.9-point gap over Self-RAG at a shared backbone, retriever, dataset and metric stands.

Two cases carry the argument and are developed in full in Appendix C: RobustRAG reporting "reducing attack success rates from over 90% to approximately 10%" beside PoisonedRAG's 0.97, which transports no number and whose certificate binds where PoisonedRAG reports 0.48; and a budget-matched re-reading that would turn PoisonedRAG's 3.7-point lead over Phantom into a 45.3-point deficit, which we decline to report because the two sides count different budgets, query populations and objectives, leaving C4 unmatched with no common unit. By our own rule that comparison is undefined, and an undefined comparison may not be replaced by a signed substitute.

5 What a Census Can and Cannot Decide

The worked ledger shows the failure exists but cannot say how common it is. So we built the screen an author or reviewer would want—one that finds candidate comparisons and adjudicates them—and measured how much of it works.

Where the corpus comes from, and how it was screened. The frame is the object of study, so its provenance is reported rather than asserted (Appendix Figure 1). 55 declared arXiv queries plus five venue and repository sources, at a fixed snapshot (2026-06-03), returned 1,904 records; 1,445 had an open PDF, 1,399 survived title-and-identifier de-duplication, 536 were retained by a released relevance pass, and 259 extracted to text with a line structure a window screen can index. Inclusion and exclusion criteria, the de-duplication rule and the per-stage actor are released. One edge is weak and the figure marks it weak: per-query hit counts were not logged at the snapshot and live indices drift, so the first stage is documented but not re-executable, while everything from the 259 released texts downward is. Hence the scope limit we hold to throughout: this frame supports composition statements about itself and about the re-weighted window universe, and none about the robust-RAG literature.

A deterministic screen over those texts marks a window as a candidate when it names a gazetteer system absent from that paper's own title and abstract region, carries at least two values formatted as results, and matches one of three declared strata: baseline-table language, explicit attribution of a number to another paper, or a survey-style score table. Nothing in the screen is a judgment; the strata are document features, never the audit outcome, and the gazetteer, cues and number pattern are released. It returns 1,703 candidates in 171 papers, and this frame—not a selection from

Judgment	AC_1	κ	$\kappa_{\max}$	$\kappa/\kappa_{\max}$	95% CI
Descriptive: reading the comparison					
How the number was obtained	0.79	+0.55	–	–	[0.13, 0.70]
Inferential: adjudicating the comparison					
Does the comparison survive?	0.22	+0.06	0.46	0.13	[-0.01, 0.11]
The proposed instrument: the five coordinates					
C1 evidence state	0.46	+0.19	0.72	0.26	[0.06, 0.29]
C2 origin / capability	0.38	+0.11	0.61	0.18	[0.04, 0.16]
C3 intervention	0.72	-0.11	0.42	-0.26	[-0.13, -0.07]
C4 outcome / unit	0.66	+0.08	0.53	0.16	[-0.05, 0.22]
C5 policy / utility	0.38	+0.10	0.51	0.19	[-0.00, 0.20]

Table 1: The census. Three judges from three developers code every candidate independently. AC_1 (Gwet, 2008) tolerates skewed marginals but rewards prevalence; κ punishes it; $\kappa_{\max}$ is the largest κ these judges could reach without changing how often each uses each label (Umesh et al., 1989; Sim and Wright, 2005), so the fourth column says how much room κ had. C3 is the one row whose κ is uninterpretable: its label is near-constant, which is the intervention exclusion of §3 showing up as a statistic. Every other coordinate had room and did not use it. Intervals resample *source papers*, since windows nest in papers and one survey can contribute dozens.

it—is what we code. Three judges from three developers—`gpt-5-mini`, `claude-opus-4.6` and `gemini-3.1-pro-preview`—code every candidate independently from the same passage plus ± 45 lines of context, the same codebook and a seed-shuffled order, never seeing one another's output; 1,690 returned a parseable coding from all three. Each returns a rationale first, then eligibility, the two sides, a label per coordinate, a verdict, a matched number, and a verbatim evidence quote we check mechanically. A second pass asks of every eligible candidate whether the other paper's number was *transported* here or *re-implemented* inside this harness. Labels combine by mechanical majority. The codebook was revised once on a declared 133-candidate development pass; 104 of those were also coded by all three judges in the deployed pass and are removed, leaving the 1,586 held-out candidates used below.

Two judgments, one recovered. 704 candidates from 103 papers are eligible by majority, and Table 1 reports every field under AC_1, Fleiss' κ, and the largest κ those judges could have reached without changing how often each uses each label ($\kappa_{\max}$; Umesh et al., 1989; Sim and Wright, 2005). The third column is what makes the first two readable: AC_1 rewards prevalence, κ punishes it, and a low κ is evidence of disagreement only where κ had room to be high. One judgment is reliable under all three: *how the other number was obtained* reaches AC_1 0.79 and κ 0.55. The inferential judgment is reliable under none: *does the comparison survive* reaches AC_1 0.22 and κ 0.06, which is 13% of its own ceiling of 0.46. The difference is +0.57 [0.50, 0.68] under AC_1 and +0.49 [0.11, 0.63] under κ, on the same items by the same judges, every interval resampling *source papers*. We do not claim the two judgments differ in *kind* rather than in difficulty—provenance is also the simpler question—only that one is recoverable at this scale and the other is not. The same ceiling makes the coordinate rows readable individually rather than uniformly: C3 alone is uninterpretable, at 87.1% one label against a ceiling of 0.42, whereas C4 is 81.4% one label but has room (0.53), so its κ of -0.11 is not a prevalence artefact (Appendix Table 9).

Did the panel execute its own rule, and did it quote honestly? §3 states a rule over C1, C2, C4 and C5; the census reports the verdict judges *stated*. Composing the rule mechanically from each judge's own coordinate labels, judges state what their coordinates imply on 0.90 of codings, and 60 of 337 majority-*survives* verdicts (18%) are contradicted by the panel's own coordinates. That is a real internal-validity defect, and it is not an alternative explanation of the negative result: agreement on the *composed* verdict is κ 0.09 against 0.06 stated, a difference of +0.02 [0.00, 0.05]. Separately, every coding must quote the passage it read, and 87% of quotes are located verbatim. Classifying all 1,879 non-empty quotes shows what the rest are: 80 are verbatim after Unicode and de-hyphenation normalisation, 76 abridge with an ellipsis whose fragments are verbatim, 59 reorder tokens present in the passage, which is what reading a linearised two-column table produces, and only 21 (1.1%) are absent from the source document. So 98.7% of quotes are supported by the passage shown, and the fabrication rate of this instrument is 1.1%, not 13% (Appendix Table 8).

Is the provenance label right, and not merely agreed on? Agreement is not accuracy, so the label must be scored against a reference. The submitted version scored it against constructed excerpts whose provenance was fixed when each was written, and reported 0.945. That number should not have been used for this purpose: across all six judges and all items, every transported item was called transported and every re-implemented item re-implemented, so the benchmark had *zero*

measured error on the two classes the composition claim depends on, and the bias correction it licensed used an identity confusion matrix and returned its input unchanged. We keep the constructed result as what it is—evidence that the distinction is not a phrase-matching artefact, since accuracy is 1.000 on items carrying no declaration sentence—and replace the validation.

The replacement is a reference on *real* windows. We draw 180 windows from the 704 the panel called eligible, stratified only on the frame stratum and on whether the cited paper's full text is held locally—a document feature and a corpus property, never a label or an outcome—giving 90 windows for which both papers are available. Four adjudicators from three developers then decide each window from the *entire* citing paper and, where available, the entire cited paper, and must support every answer with a quote we locate mechanically in the document that adjudicator was shown; at the both-texts tier, one quote in each paper. A reference label is established where a majority of quote-verified adjudicators agree, and windows without one are reported rather than dropped. This reference is not human gold and we make no claim that it is; it is *evidence-privileged*, better informed than the instrument it scores by a documented margin, and auditable end to end.

Against it (Appendix Table 7) the deployed panel recovers provenance at 0.91 [0.86, 0.96] on 170 windows, against a majority-class baseline of 0.79; recall is 0.80 on transported windows and 0.93 on re-implemented ones. The measured confusion is not the identity—sensitivity 1.00 and specificity 0.99 for the transported class—and moves the re-implemented share of the frame to 88.1% against 87% uncorrected. Removing the adjudicator sharing a checkpoint with the deployed panel leaves accuracy at 0.89. The same reference scores the verdict far less kindly, which is the sharpest form of the paper's negative result: 0.82 [0.74, 0.89] against a majority-class baseline of 0.77 on 114 windows. Three frontier judges reading a published passage recover the adjudication barely better than a constant predictor. The reference is also harder to establish here—a quote-verified majority exists on 129 of 180 windows against 171—so the disagreement is not confined to weak judges reading short passages.

What the frame is made of, and does that survive the screen? Of the 658 eligible candidates all three judges labelled, 610 are assigned a provenance class; 533 of those are re-implementations—the citing paper re-ran the other system in its own corpus, retriever, generator and metric—and 77 transport a number across a harness boundary. Inside this frame, cross-paper comparison in robust RAG is therefore mostly a within-harness activity, which is this paper's most useful positive fact: the incommensurability of §4 has a narrow, identifiable habitat that an automated screen can find even though it cannot adjudicate what is inside it. We report no verdict-based failure rate, because a rate computed from a label whose κ is 0.06 is the same non-evidence as an estimate.

That composition might be an artefact of a screen whose first stratum is baseline-table language, since baseline tables are where re-implementations live. Against a universe of 6,710 windows naming another paper's system and carrying a result-formatted value, 420 of the 5,007 the screen missed were coded by the same judges under the same prompts (Appendix Table 15). Missed windows are eligible far less often (0.34 against 0.44), and recall is 0.50 [0.26, 1.00] for transported juxtapositions against 0.28 [0.24, 0.33] for re-implemented ones, so the screen surfaces a transported juxtaposition more readily, not less. Re-weighting moves the re-implemented share *up*, from 87% to 93% [86, 96]. That second figure turns on 7 sampled windows, and we say so where we use it: reassigning all of them moves it to 96.1%, doubling their count to 88.8%, and a paper-clustered bootstrap gives [0.863, 0.961]. The direction survives every perturbation the data admit and the magnitude does not, so we report both figures and claim neither for the literature.

6 What Exactly Breaks

The census cannot separate three explanations of its own negative result: the adjudication task is underdetermined by published passages, the schema cannot be coded, or the judges cannot report underdetermination even when they detect it. Observational agreement cannot separate them. So we intervene three times, holding the judges, the codebook and the items fixed and moving one thing at a time: the evidence available, the answer set available, and the passage itself. The three answers are different, and together they say something narrower and better supported than the submitted version claimed.

Intervention 1: give the judges the whole paper. If the census fails because a 45-line window does not disclose the coordinates, then disclosing them should repair it, and the strongest available disclosure is the papers themselves. The same three judges re-code 180 sampled windows under the same codebook, once from the entire citing paper and once—on the 90 windows where we hold it—from both papers in full (Table 2). That is roughly a hundredfold increase in evidence, including the setup sections that state how baseline numbers were obtained.

It helps, measurably, and it is not enough. Verdict agreement rises from κ 0.06 at the window to 0.29 with the citing paper in full, a paired difference of +0.20 [0.09, 0.36] resampling source papers; adding the cited paper gives 0.16, within noise of the citing paper alone on the windows both cover. So the census figure of 0.06 is in large part an artefact of what a screen can see, and this is the first interventional evidence here that disclosure would help a real reader. But a fivefold rise lands at *fair*, not reproducible: the same instrument reaches κ 0.74 on constructed passages that state their coordinates, and no amount of the actual papers closes that distance.

The provenance arm is the positive control that makes this reading defensible. The identical intervention takes provenance from κ 0.68 to 0.92 and 0.97 (+0.36 [0.11, 0.80]), which is near-perfect, because whether a number was transported *is* settled by comparing the two documents. The ladder does what a manipulation should: it saturates the judgment whose answer the added evidence contains, and moves but does not resolve the judgment whose answer it does not.

Intervention 2: let the judges say the passage does not say. The remaining internal explanation is the instrument. The deployed codebook offers *survives*, *narrows* and *undefined*, and *undefined* means a disclosed, unrepaired mismatch—so a judge detecting silence has nowhere to put it, and this paper proves that mechanism on a neighbouring label, where a fourth provenance option moved the asserted-no-number class from 0% to full recovery. We therefore add a fourth verdict, `not_determinable_from_text`, described in the same register as the other three and preferred under uncertainty, and change nothing else.

Judges use it, on 42.4% of codings and by every judge (29% to 53%), and their *survives* rate falls by -0.22, so the deployed instrument was indeed suppressing a report of silence. Agreement still does not follow. Collapsing the new label back onto *undefined*, so both arms live in one three-label space and a change is a change in how items are partitioned rather than a relabelling, verdict κ moves by +0.01 [-0.07, 0.11]. The judges now agree that many passages do not say; they do not agree about *which*.

Manipulation	n	Obs.	κ	95% CI	κ (com./max)
Verdict: *does the quoted gap survive?*					
$\pm$45 lines (deployed)	180	0.44	0.06	[-0.07, 0.13]	0.05
citing paper, full text	179	0.64	0.29	[0.20, 0.40]	0.24
both papers, full text	89	0.57	0.16	[0.04, 0.32]	0.16
Δ full paper $-$ window			+0.20	[0.09, 0.36]	
Provenance: *transported or re-implemented?*					
$\pm$45 lines (deployed)	166	0.86	0.68	[0.21, 0.79]	0.60
citing paper, full text	180	0.97	0.92	[0.51, 0.96]	1.00
both papers, full text	89	0.99	0.97	[0.50, 1.00]	0.97
Δ full paper $-$ window			+0.40	[0.15, 0.87]	
Abstention: an explicit `not_determinable_from_text` *verdict*					
deployed codebook	633	0.44	0.06		0.46
four-verdict codebook	633	0.41	0.13	[0.02, 0.24]	0.54
collapsed to deployed space	633	0.55	0.06		0.36
Δ collapsed $-$ deployed			+0.01	[-0.07, 0.11]	

Table 2: Two interventions, with the judges, the codebook and the items held fixed. *Top:* the evidence varies. The window row is the deployed census condition and is not re-run; the last column scores every tier on the windows all three tiers cover. Supplying the papers raises verdict agreement without making it reproducible, while driving provenance to near-perfect agreement – the positive control that makes the first reading defensible. *Bottom:* the answer set varies. Judges take the abstention option on 0.424 of codings and their *survives* rate falls by -0.215, so the deployed instrument was suppressing a report of silence; agreement does not follow. The collapsed row puts both arms in one three-label space, so a change there is a change in how items are partitioned rather than a relabelling.

Does any of this depend on the judges being proprietary? Every judge above is an API-served checkpoint, so the negative result could be a property of one class of model, and σ_r below would be estimated from a narrow capability band. We therefore add 9 open-weight checkpoints answering byte-identical prompts under greedy decoding in one harness, their weights and revisions pinned. The census result replicates and if anything strengthens: on the 305 windows all pools cover, verdict κ is 0.05 for the API pool and 0.09 for the open pool, against a ceiling of 0.46 their own marginals would have permitted (Appendix Table 19). The abstention result replicates too: open judges take the abstention option on 34.0% of cod-

ings and collapsing it moves κ by -0.04. Adjudication is not recovered by open models either, and not because they label too uniformly to agree.

Intervention 3: write passages that do state the coordinates. The third intervention is the constructed benchmark, the only one that moves the verdict decisively. Each of 432 items is a short synthetic excerpt—a setup passage describing how each side was evaluated, and a results passage placing two numbers side by side—behind which a specification fixes, for C1, C2, C4 and C5, whether the two sides agree and whether the excerpt discloses the fact, from which the verdict follows by the rule of §3. *Disclosure* has three levels: FULL states every coordinate; WITHHELD-NEUTRAL withholds one but fixes the verdict by something the excerpt does state, so all three verdict classes occur equally at a constant baseline of 0.34; and WITHHELD-DECISIVE makes the correct answer *that the passage does not say*, a degenerate arm whose constant baseline is 1.00 and which is scored against that and excluded from every pooled figure. *Presentation* is a second, paired factor: the same facts as one sentence per coordinate, and as paper-style prose. 6 judges from three developers answer every item.

On passages that state them, judges recover the coordinates at 0.95 and agree on the verdict at κ 0.74; the rule composed with those labels reaches 0.861 against a constant baseline of 0.336 on the non-degenerate items, the weakest judge included (Table 17). C3 is the exception that confirms §3: judges recover it at chance even when it is stated. Writing the same facts as prose costs a measurable but small amount—coordinate recovery 0.93 to 0.89, composed rule 0.89 to 0.83, inter-judge κ unchanged—so readability is not the binding constraint.

What fails is silence, and how much of it evidence can repair. Withholding a coordinate is not by itself what breaks adjudication. When one is withheld and something else still fixes the answer, judges lose accuracy on the *coordinate* (0.95 to 0.87) and nothing on the verdict, and agreement is if anything higher (κ 0.84 against 0.74). The collapse is specific to the arm where the right answer is that the passage does not say: verdict accuracy 0.32 and composed rule 0.33 against a constant predictor scoring 1.00, with agreement at κ 0.25.

The four interventions therefore place the census result between two bounds. About a fifth of a κ point is recoverable by supplying evidence the window omitted (+0.20 [0.09, 0.36]). None is recoverable by giving the judge somewhere to put a report of silence (+0.01 [-0.07, 0.11]), by running the rule for it (+0.02 [0.00, 0.05]), or by changing the model class (open-weight κ 0.09). What remains—the distance from 0.29 to 0.74—is what only rewriting the papers would close. The honest summary is neither that judges cannot report silence nor that evidence is irrelevant, both of which we tested and disproved: it is that these passages, even read in full, do not fix the comparison.

Two coordinates or five? The fair comparison for a five-field checklist is against the best two-coordinate *estimator* (Table 17). On the non-degenerate items the five-coordinate composed rule reaches 0.861 against 0.832, a paired difference of +0.029 with exact McNemar $p = 0.00146$, and the gap widens on the arm whose correct answer is that the passage does not say (0.33 against 0.11), because asking about C2 and C5 is what makes their absence visible. Against that stands the corpus, where the C1/C4 codebook gives verdict κ 0.34 and the five-coordinate one 0.31 (+0.03, [-0.05, +0.12]). The two settings measure different quantities—accuracy against a reference, and agreement without one—so we claim only what holds in both: C1 and C4 are the minimum any checklist can ask, and C2 and C5 earn their place by making silence detectable, not by raising inter-judge agreement on papers as they are written today.

7 What Follows for Practice

Reviewers are not interchangeable, and they are the unit. Deciding comparability at scale means running a panel of model reviewers, and that panel is not exchangeable: the nine-judge pool holds different thresholds rather than differing in noise around one, with the share of eligible windows a judge calls sound ranging from 18% to 78% on identical items. The open-weight pool makes the point sharply, because capacity is measurable there: across 9 checkpoints spanning 8B to 72B the same share runs from 23% to 79%, the two extremes sit within one parameter decade of each other, the largest checkpoint lands mid-range at 70%, and two 8B checkpoints— `qwen3-8b` and `internlm3-8b`—read identical windows at 27% and 76%. Threshold is not a function of scale. That is also why the verdict's κ ceiling is only 0.46. A separate

crossed benchmark of 600 synthetic deployment decisions answered by 28 reviewers from 13 developers reproduces the heterogeneity (I^2 98%; Appendix H). The consequence is the general form of the language-as-fixed-effect fallacy: if prompts are assigned to reviewers, reviewers remain the treatment units after majority voting, and analysing at the source inflates type-I error. The missing parameter is σ_r, and the objection to the submitted estimate was that six judges cannot determine it: its interval spanned a factor of 4.0. The fix is more reviewers, so we estimate it on 22 judges—13 API-served and 9 open-weight—all answering the constructed benchmark, whose reference verdict is fixed before any judge sees the item. That gives $\hat{\sigma}_r = 0.50$ [0.33, 0.62], an interval spanning a factor of 1.9 rather than 4.0. It also shows the narrow pool was biased downward, not merely imprecise: the API judges alone give 0.37 because they occupy a narrow capability band, and adding open checkpoints that reach lower accuracies raises it. Five public crossed annotation corpora sit inside the same envelope at 0.31–0.78 (Appendix Table 20). We draw only the conclusion holding everywhere in it: at $\sigma_r = 0$ a source-level analysis is calibrated (4.7% at 80 sources), and at every value the anchors admit it rejects a true null 8% to 28% of the time, while two retrospectively executable reviewer-level analyses hold the nominal level throughout (Appendix G).

A pre-collection checklist. Four decisions follow (Appendix Table 35), and we state what supports each. *State the evidence state and the outcome unit* (C1, C4): their absence is what makes adjudication undecidable rather than merely hard. The support is interventional—supplying the papers in full raises verdict agreement by +0.20 [0.09, 0.36]—not observational, since windows that merely *mention* more coordinate terms do not yield more reproducible verdicts (-0.08 [-0.27, 0.05]). Stating a coordinate means stating both sides' values for it, not naming it. *State the capability and the action set* (C2, C5): they make a silence detectable, and every security case in §4 is unmatched on C2. *Say whether a quoted number was transported or re-implemented*: it is the one property an automated audit recovers against a reference on real windows (0.91 [0.86, 0.96]), and the one the evidence ladder drives to near-perfect agreement. *Draw the unit map and power it*: release per-reviewer pre-adjudication votes so the analysis can be redone at the right unit, and treat a twelve-reviewer two-arm design as a pilot (Card et al., 2020).

What we do *not* recommend is deploying this audit as an adjudicator. Against the privileged reference the deployed panel's verdict accuracy is 0.82 against a majority-class baseline of 0.77. What it recovers is the comparison and its provenance, and that is what an automated screen should hand a reviewer.

8 Conclusion

In the frame this paper documents and screens, cross-paper comparison in robust RAG is mostly not what it appears to be: of the 610 candidates assigned a provenance class, 533 re-implement the other system inside the citing paper's own harness. That distinction is what automation recovers—κ 0.55 between judges, 0.91 [0.86, 0.96] against a reference built by reading both papers in full—and it locates the narrow stratum in which a quoted gap can be undefined rather than imprecise.

Adjudicating those gaps is not recovered, and four interventions decompose why. Supplying the entire citing paper raises verdict agreement by +0.20 [0.09, 0.36], so a large share of the census figure is an artefact of what a 45-line screen can see. An explicit *the passage does not say* option, taken on 42.4% of codings, moves it by +0.01 [-0.07, 0.11]; composing the rule from the judges' own coordinates moves it by +0.02 [0.00, 0.05]; and 9 open-weight checkpoints reproduce the failure at κ 0.09. Neither the option set, nor rule execution, nor the model class is the obstacle. What is left is the distance from 0.29, with both papers read end to end, to the κ 0.74 the same instrument reaches on prose written to state its coordinates. That distance is a fact about how the field writes rather than about how models read, and it is why the coordinates belong in a disclosure requirement rather than in a score—and why what a screen should hand a reviewer is the comparison and its provenance, not the verdict.

Limitations

What this paper supersedes. Three results in an earlier version of this work do not survive and are replaced rather than hedged, and we group the accounting here instead of narrating it through the argument. A verdict-based failure rate for transported comparisons has been removed, because a

direction inferred from a label whose κ is 0.06 is not weaker evidence than an estimate but the same non-evidence. A constructed provenance benchmark that reported 0.945 has been demoted from validation to a lexical control, because every judge was perfect on the two classes the composition claim depends on, so it had no power to detect the bias it was invoked to rule out and the correction it licensed was the identity map; §5 replaces it with a reference on real windows. And an earlier disclosure benchmark scored a single-class arm against a three-class baseline; §6 separates the three cases it confounded and scores every arm against its own constant predictor.

The frame. The twelve worked comparisons are cases, not a census; both sides had to be present in local full text with an explicit number, which favors open-access and widely cited work. Our coding is ours, and the independent re-coding in §4 reproduces it on the binary distinction we use but on only 58% of three-way labels, which is why no numerical rate from the ledger appears in this paper. The census removes author selection but introduces its own limits. Its first stage is documented and not re-executable: per-query hit counts were not logged at the snapshot date, so a third party can reproduce the frame from the 259 released texts downward but cannot reconstruct the population those texts were drawn from. Every composition figure is stated for the frame or for the re-weighted universe of 6,710 windows, never for the field, and the re-weighted figure turns on 7 sampled windows, which is why §5 prints that count beside it and reports the perturbation range 88.8%–96.1%.

The reference. The privileged reference is not human gold and is not a substitute for one. Its adjudicators are language models; what makes it a reference is that they read both papers where the instrument read 45 lines, and that every vote counted toward it carries a mechanically located quote. Two of its four adjudicators share a developer and one shares a checkpoint with the deployed panel, because those are the strongest checkpoints those developers serve, so independence is by evidence and by verification rather than by model identity; the leave-that-judge-out column exists for this reason. It is established on 171 of 180 windows for provenance and 129 for the verdict, and windows without a quote-verified majority are reported, not dropped. Nothing here establishes what a panel of people would do.

The interventions. The evidence ladder shows that supplying the full papers raises verdict agreement without making it reproducible. That is evidence about how much of the census figure is an evidence deficit and how much is not; it is not proof that the residue is underdetermination, since a judge may fail to use evidence it is given, and 180 windows from 63 papers is a small design. We report the provenance arm of the same ladder as the positive control that makes the reading defensible, since the identical intervention drives that judgment to near-perfect agreement. The abstention arm changes two things at once by necessity, an option and an instruction to prefer it, so it identifies their joint effect. Neither is a randomized experiment on disclosure, which cannot be run on published text; the constructed benchmark is the closest available substitute and is synthetic by construction.

Two of the ladder's own numbers should temper it. Its both-texts tier covers only 90 windows, so its interval is wide and it is not distinguishable from the citing-paper tier. And even the privileged adjudicators fail to reach a quote-verified majority on the verdict for 51 of 180 windows, against 9 for provenance: the disagreement this paper measures is not confined to weak judges reading short passages.

The instruments. The tier control identifies capability tier against developer but not against everything else: "frontier" and "small" are the developers' own tiering, and the OpenAI cells are served over the vendor CLI rather than the local proxy, because the proxy injects a parameter those checkpoints reject—the same prompt and codebook, but not an identical harness. The exact randomizations are valid at the judge level and they are small. The constructed benchmark is templated prose with a synthetic specification, and one limit is structural: each item's reference verdict *is* the decision rule applied to that item's coordinate values, so the rule cannot be wrong on it by construction. What its accuracy measures is whether a judge can read coordinates off prose, propagated through a deterministic function. We claim no field-level rate from it, since its classes are balanced by construction. The five-versus-two comparison is reported in both directions because the two settings disagree, and neither result alone settles what a checklist should ask for.

The parameter. The reviewer-heterogeneity estimate now rests on 22 judges rather than six, which narrows its interval to a factor of 1.9, but they are all model judges and the five public annotation corpora are not comparability adjudication. The census anchor's reference is a leave-one-out majority of judges who agree at κ 0.06, so it re-expresses threshold dispersion as much as ability variance; the constructed benchmark's reference is fixed before any judge sees the item and is the one we use. Which checkpoints enter the pool still moves the estimate—0.37 for the API judges alone against 0.50 pooled—so σ_r is a property of a panel a designer assembles, not a constant of the task, and we report the envelope 0.12–0.75 with only the conclusion that holds across it. That the wrong-unit inflation exists is not new (Clark, 1973; Judd et al., 2012); what is new is the parameter for this task, and it is a model-panel parameter with a wide interval. The simulation uses binary correctness, a common wrong-label state, balanced three-of-six panels and a logit random-effects model; it demonstrates a design failure mode and the analyses that repair it, not a prevalence.

The crossed deployment benchmark is templated arithmetic prose with a synthetic loss rule; its model sample spans 13 developers but remains a convenience sample, and its capability and utility effects are null, so we draw only the reviewer-heterogeneity conclusion from it. Every judgment in this paper is produced by a model or by a deterministic rule. Local hashes authenticate bytes, not behavioral independence or external preregistration.

Ethics Statement

This paper contains no human-subjects research. No annotation was collected from any person, no participant was recruited or contracted, and no result reported here derives from labels we gathered from people. Every judgment in the census, the tier-control pool, the codebook ablation, the ledger re-coding, the privileged-information reference, the evidence ladder, the abstention arm, the two constructed benchmarks, the out-of-frame recall sample and the deployment panel is produced by a language model under a released prompt with released raw outputs, or by a deterministic rule over released artifacts. We adopted this design deliberately. A paper arguing that an unauditable declaration is not evidence should not rest a section on one, and a fully machine-executed pipeline is inspectable end to end by anyone who downloads it.

This choice has a cost we state rather than gloss. The privileged reference in §5 is a reference by virtue of evidence and mechanical verification, not by virtue of being human, and readers should not substitute it for a human standard; where the manuscript says a judgment is "recovered" it means recovered against that reference and against inter-judge agreement, and nothing more. Model judgments are not human gold, so the agreement decomposition in §5 is a statement about automated auditing and not about people, and the checklist in §7 is addressed to authors who will run such panels rather than offered as a claim about human review.

The published-paper audit uses public scholarly documents and introduces no participant contact. The comparability cases quote short passages from published papers for verification purposes only, and every such quotation is checked verbatim against its cited line range by the released verifier. The five public annotation corpora used as external reference points in Table 20 are re-analysed from their authors' public releases under their stated licences; we contact no annotator, re-identify no one, and report only aggregate variance components.

Two limits of this choice should be stated rather than glossed. The constructed benchmark contains no participant data and no third-party quotation, so its realism is bounded by our own templates; and the privileged reference quotes short passages from published papers for verification purposes only, every such quotation being checked verbatim against its cited line range by the released verifier. The reported open-model runs use deterministic decoding and release raw outputs. No attack payloads or poisoned corpora are released.

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A Substantive Comparability in Robust RAG

We use the literature map as a source of boundary cases, not as prevalence evidence. Five *operating questions* organise discovery: how evidence is constructed, how a model uses it, how deliberate corruption changes the game, how signals become actions, and how those actions are evaluated. They are a different cut of the same literature from the four failure surfaces of §3, and both are different again from the five *reporting coordinates* of §3, which describe not a literature but a single proposed contrast. The boundary cases below are what made the coordinates necessary.

Evidence availability is not evidence use. Irrelevant-context studies test whether retrieved distractors redirect a generator (Yoran et al., 2024); noise-aware training changes the generator's training distribution (Fang et al., 2024); RGB supplies diagnostic stress conditions (Chen et al., 2024a); and Chain-of-Note changes how retrieved documents are read (Yu et al., 2023). A filter, a trained reader, and a benchmark may share a harness, but their native scores do not rank interchangeable interventions. Insufficient evidence creates another boundary: when no passage can support an answer, re-retrieval or abstention can dominate better distractor resistance (Joren et al., 2025).

Support, authority, conflict, and attack are distinct. RAGTruth annotates unsupported spans, while evaluators estimate claim or response support (Niu et al., 2024; Es et al., 2024; Saad-Falcon et al., 2024); knowledge-conflict methods instead study competition between parametric memory and retrieved evidence (Jin et al., 2024b; Zhang et al., 2025b; Wang et al., 2025a). A response can be correct but unsupported, or faithful to a weak source. Provenance and timestamp determine which source should govern (Petroni et al., 2021; Vu et al., 2024; Kasai et al., 2023). Deliberate corruption changes the game again: corpus poisoning, indirect prompt injection, and backdoors differ in corpus write access, query knowledge, retriever visibility, insertion budget, persistence, and objective (Zou et al., 2025; Chaturvedi et al., 2025; Chaudhari et al., 2024), and their defenses prevent, detect, certify, trace, quarantine, or recover under different assumptions (Xiang et al., 2024; Si et al., 2025; Zhang et al., 2025a). An attack-success comparison is transportable only within a matched capability envelope.

A signal is not a policy. Calibration predicts a property of a score (Guo et al., 2017); selective prediction trades coverage against risk (Geifman and El-Yaniv, 2017); sufficient-context classification asks whether available passages can support an answer; adaptive retrieval chooses whether to search, verify, answer, or abstain (Jang et al., 2025; Joren et al., 2025; Labruna et al., 2024). Self-RAG, CRAG, and CtrlA expose this policy layer explicitly (Asai et al., 2024; Yan et al., 2024; Liu et al., 2025b). Comparing them requires a common action set, budget, and loss: accuracy alone cannot price extra retrieval, coverage, or refusal.

Comparison ladder. The strongest contrast reruns alternatives with the same corpus snapshot, retriever, generator, evidence state, capability envelope, action set, and outcome. When that is impossible, synthesis should retain only the common scope supported by both studies. If a decision-critical field is absent, the comparison is underspecified rather than evidence that either method is better. Appendix B gives paper-level contrasts and the documentary-map boundary; the next section turns these recurring boundaries into a claim-level audit.

B Extended Synthesis by Operating Question

Sections 4–4 organise robust RAG by failure surface, which is where a paper sits. This appendix cuts the same literature the other way, by the *operating question* a study answers, and develops the paper-level contrasts the main text compresses. It is a synthesis, not an exhaustive bibliography: the released 342-row evidence map holds the long-tail records, source strata and intervention labels.

Two limits carry over from §5 and govern how the contrasts below may be read. The screening flow's first stage is documented but not re-executable, so this synthesis identifies recurring scientific distinctions rather than estimating prevalence or ranking evidence families; and the map carries no study-level risk-of-bias weighting. A review-directed recall check also surfaced older, pre-freeze publications absent from the frozen map, which we cite as qualitative context and log as post-freeze discoveries without entering them into any denominator. Where this survey states a number rather than a distinction, it does so only for the frame of §5.

B.1 When Does Retrieved Context Help?

Irrelevant-context and noise-aware-training studies intervene at different points. Yoran et al. (2024) shows that retrieved distractors can actively redirect generation and tests filtering and targeted fine-tuning. Fang et al. (2024) instead trains a generator across random, relevant, and counterfactual retrieval conditions. RGB supplies stress conditions and diagnostic abilities rather than a competing mitigation (Chen et al., 2024a). These papers can share an end-to-end harness, but their native results do not rank a filter, a trained reader, and a benchmark as interchangeable objects.

Context construction creates a second boundary. Chain-of-Note changes how the generator reads retrieved documents, whereas compression and reranking alter what reaches it (Yu et al., 2023; Wang et al., 2023; Trappolini et al., 2025). Long-context studies show that retaining more passages can improve evidence availability while worsening position sensitivity or distraction (Jin et al., 2024a; Gabín et al., 2026). A fair comparison therefore fixes the candidate evidence, generator, decoding, and budget when the intended claim is about the intervention rather than the entire system.

Insufficient evidence is not merely severe distraction. Sufficient Context labels whether the available passages can support an answer and evaluates selective behavior (Joren et al., 2025). A reader that resists a hard negative and a controller that abstains when support is absent solve different problems. Reports should state both answer-bearing support availability and the scored action.

B.2 Which Knowledge Should Govern an Answer?

Knowledge-conflict work separates context–memory disagreement from conflict among retrieved sources. Tug-of-War measures how models trade parametric and contextual knowledge; FaithfulRAG models fact-level discrepancies; Astute RAG reconciles imperfect retrieval with internal knowledge (Jin et al., 2024b; Zhang et al., 2025b; Wang et al., 2025a). Although all three concern "conflict," their experimental units, source assumptions, and desired responses differ. The appropriate outcome may be answer correctness, contextual faithfulness, calibrated disclosure, or source reconciliation. Earlier entity-substitution experiments and PopQA isolate when parametric memory resists or benefits from retrieved evidence (Longpre et al., 2021; Mallen et al., 2023). ConflictBank broadens the benchmark unit to millions of claim–evidence pairs across misinformation, temporal conflict, and semantic ambiguity (Su et al., 2024); its scale does not make those conflict types interchangeable.

Grounding resources address another layer. RAGTruth provides span-level hallucination labels, while RAGAS and ARES estimate response or context support (Niu et al., 2024; Es et al., 2024; Saad-Falcon et al., 2024). These evaluators can identify an unsupported answer, but support alone does not establish that the cited source is current or authoritative. A system can be faithful to weak evidence or correct for a reason absent from its citations.

Source reliability and time therefore belong in the evidence record. Recent work models source reliability directly (Hwang et al., 2025), and other studies examine source preference and contextual authority (Schuster et al., 2026; Li et al., 2026). The comparison-relevant question is not only whether the model used retrieved text, but whether the benchmark specified which source should govern, at what timestamp, and under what corroboration rule. KILT standardizes provenance against a fixed knowledge source, while FreshQA and RealTimeQA test knowledge whose answer changes over time (Petroni et al., 2021; Vu et al., 2024; Kasai et al., 2023). ALCE and generative-search verifiability work separately score citation attribution and whether cited sources actually support generated statements (Gao et al., 2023; Liu et al., 2023).

Multi-agent methods add yet another intervention boundary. MAIN-RAG uses agent voting to filter retrieved documents, MADAM-RAG assigns agents to conflicting documents before aggregation, and DRAG introduces debate during retrieval and generation (Chang et al., 2025; Wang et al., 2025b; Hu et al., 2025). General multi-agent debate supplies methodological background (Du et al., 2024), but these RAG systems should be compared only when the evidence allocation, agent budget, aggregation rule, and target outcome match.

B.3 What Changes Under an Attacker?

Corpus poisoning makes evidence unreliability intentional. PoisonedRAG inserts passages to steer target answers; earlier corpus-poisoning work manipulates retrieval through adversarial passages (Zou et al., 2025; Zhong et al., 2023). Attack success is conditional on corpus write access, insertion

budget, query knowledge, retriever visibility, and target scope. Omitting these fields can make two attack rates numerically similar but scientifically incomparable.

Indirect prompt injection and backdoors target a different interface. Retrieved text may contain instructions that override the user task, or a trigger may activate behavior learned during training (Chaturvedi et al., 2025; Chaudhari et al., 2024; Cheng et al., 2024). These attacks should not inherit claims from factual poisoning merely because both place malicious text in the context. Retrieval realism, instruction-following behavior, and trigger access define distinct games.

Defenses likewise have different roles. Certified RAG gives guarantees under a bounded corruption model; filtering and semantic isolation demote suspicious documents; traceback identifies likely poisoned sources (Xiang et al., 2024; Si et al., 2025; Zhang et al., 2025a). Prevention, certified accuracy, detection, and recovery require different outcomes. Any cross-defense comparison should pair the same capability envelope with clean utility and state whether the defense prevents, detects, traces, quarantines, or recovers.

B.4 How Does a Signal Become an Action?

Calibration, answerability, and retrieval usefulness are related but not identical signals. Calibration-oriented RAG predicts reliability, whereas Sufficient Context asks whether the retrieved evidence can support an answer (Jang et al., 2025; Joren et al., 2025). A well-calibrated score can still feed a poor threshold or action policy, and an insufficient-context detector does not determine whether retrieval, clarification, or abstention has the highest utility.

Adaptive systems make the policy explicit. Self-RAG learns when to retrieve and critique; CRAG adds corrective retrieval; CtrlA separates uncertainty sources to select control actions (Asai et al., 2024; Yan et al., 2024; Liu et al., 2025b). Their reported gains depend on the available action set and resource budget. Answer-only accuracy cannot capture the benefit or cost of extra retrieval, verification, or refusal.

Selective evaluation should therefore retain risk, coverage, action, and cost rather than collapse them into one score. Trust-or-abstain methods, conformal controllers, and uncertainty benchmarks instantiate different operating points (Zhu et al., 2026; Chen et al., 2024b; Nguyen et al., 2026). Comparisons are strongest when systems share evidence and actions and are evaluated across a risk–coverage or utility curve.

B.5 What Can the Evidence Map Establish?

The map documents that robust-RAG research spans evidence construction, evidence use, adversarial integrity, and action control. Existing surveys provide architectural, evaluation, trustworthiness, security, conflict, and abstention views (Gao et al., 2024; Yu et al., 2024; Zhou et al., 2024; Xu et al., 2026, 2024; Wen et al., 2025). Controlled searches for RAG best practices provide an empirical component comparison under shared tasks (Wang et al., 2024), complementary to survey taxonomies and to the claim-level audit studied here. Table 30 states the narrower claim-level increment tested here.

The map cannot establish that these families occur at representative rates, that its taxonomy is complete, or that one family has stronger evidence. Accepted-paper additions strengthen several historical lines, while 205 of 342 mapped rows remain preprints against 137 peer-reviewed, and coding depth varies across legacy strata. Map counts therefore provide documentary context only. Every measured claim in this survey rests elsewhere: the taxonomy test on the constructed benchmark of §6, the composition figures on the screened frame of §5, and the design claim on the assignment-unit simulation.

C Comparability Case Ledger

Case	Src	Mismatch	Verdict
SEC-01	rerun	C1, C2, C4	undefined
SEC-02	invited	C1, C2, C4	undefined
SEC-03	no num.	C1, C2, C4	undefined
SEC-04	quoted	C2, C4, C5	undefined
SEC-05	rerun	C1, C2	narrows
NOISE-01	rerun	C3	survives
NOISE-02	rerun	C1, C3	narrows
NOISE-03	invited	C1, C4	undefined
NOISE-04	invited	C1, C3, C4	undefined
NOISE-05	rerun	C1, C5	survives
FAITH-02	quoted	C1, C4	undefined
CALIB-03	invited	C1, C4	narrows
FAITH-01	control	C4	narrows
CALIB-01	control	C3	survives
CALIB-02	control	C3	survives

Table 3: Verdicts on the worked ledger. *Src*: whether a paper states the comparison or a survey invites it by grouping. C1 evidence state, C2 origin/capability, C3 intervention, C4 outcome/unit, C5 policy/utility. *Src* also records how the other side's number reached the citing paper: *quoted* transports it, *rerun* re-implements the other system in this harness, *no num.* asserts a relation without giving the other side's number, and *invited* is a juxtaposition no author stated. The rule of §3 is applied exactly: only C1, C2, C4 and C5 threaten a comparison, and a C3 mismatch never does. The table has 12 cross-paper rows—8 stated by an author and 4 invited by a survey's grouping—above the rule, and three *control* rows below it in which both sides come from one harness and no cross-paper transport occurs. An independent 6-judge panel reproduces these verdicts on 100% of the cross-paper rows for the binary distinction we use and 58% for the three-way label (Table 4), which is why no rate from this table is quoted as a result. Numbers, coordinate-matched re-readings and consequences are in Appendix Table 5; all 102 provenance, numeral and verbatim-quotation checks pass.

Case	Kind	Ours	Independent panel	Agree
CALIB-03	cross	narrows	unde 6	0/6
FAITH-02	cross	undefined	surv 3, unde 3	3/6
NOISE-01	cross	survives	surv 6	6/6
NOISE-02	cross	narrows	unde 6	0/6
NOISE-03	cross	undefined	unde 6	6/6
NOISE-04	cross	undefined	unde 6	6/6
NOISE-05	cross	survives	surv 4, unde 2	4/6
SEC-01	cross	undefined	unde 5, surv 1	5/6
SEC-02	cross	undefined	unde 5, narr 1	5/6
SEC-03	cross	undefined	narr 5, unde 1	1/6
SEC-04	cross	undefined	unde 6	6/6
SEC-05	cross	narrows	unde 3, surv 3	0/6
CALIB-01	ctrl	survives	surv 3, unde 3	3/6
CALIB-02	ctrl	survives	surv 6	6/6
FAITH-01	ctrl	narrows	unde 6	0/6

Table 4: The ledger holds up on the distinction it is used for and not on the three-way label. Six judges from three developers re-code every case from the neutral statement of the juxtaposition and the verbatim source excerpts alone, never seeing our coordinate labels, our reading, our matched number or our verdict. On the 12 cross-paper cases the panel majority reproduces our three-way verdict on 58% of cases ($\kappa = 0.32$), and agreement among the six judges themselves is $\kappa = 0.52$. On the binary distinction the paper's conclusion actually uses—does the quoted gap stand as reported—the majority reproduces us on 100% and the panel is unanimous on 67% of cases. This is why we report the ledger's negative statements and not a rate.

ID	Src	Compared results	Quoted gap	Mismatch	Verdict	Consequence of matching the coordinate
SEC-01	rerun	PoisonedRAG vs. CorpusPoisoning	1% vs 84.2% for the same attack	C1, C2, C4	undefined	C4 differs: 0.01 scores answer manipulation, 84.2 scores top-20 retrieval, so no gap is defined.
SEC-02	invited	PoisonedRAG vs. Phantom	+3.7% ASR in favor of PoisonedRAG	C1, C2, C4	undefined	No matched operating point exists for C4 (targeted answer vs refusal), so no gap is defined at any budget.
SEC-03	no num.	RobustRAG vs. PoisonedRAG	Defense credited with reducing poisoning ASR to <=15%	C1, C2, C4	undefined	No number is transported; RobustRAG re-implements the attack, and its own poisoning ASRs are 37-86.
SEC-04	quoted	BadRAG vs. PoisonedRAG	+1.2% in favor of BadRAG (98.2% vs 97%)	C2, C4, C5	undefined	Objectives differ (denial of service vs targeted answer), so the 1.2-point gap does not rank the two attacks.
SEC-05	rerun	AIP vs. PoisonedRAG	+66.66% ASR in favor of AIP	C1, C2	narrows	At five injected documents on MedSquad the 66.66-point gap stands; it does not transfer to NQ.
NOISE-01	rerun	CRAG (Self-CRAG) vs. Self-RAG 7B	+6.9 accuracy in favor of Self-CRAG	C3	survives	All five coordinates match, so the 6.9-point gap stands as reported.
NOISE-02	rerun	CtrlA 7B vs. Self-RAG 7B	+6.9 accuracy in favor of CtrlA	C1, C3	narrows	Backbone and retriever differ, so the 6.9-point gap has no determined sign.
NOISE-03	invited	RGB (ChatGPT) vs. RAAT (ChatGPT3.5)	+4.53 accuracy/EM in favor of RGB's ChatGPT	C1, C4	undefined	Substring accuracy is weaker than exact match, so the 4.53-point gap has no determined sign.
NOISE-04	invited	Chain-of-Note vs. RAAT (ChatGPT3.5)	+7.55 vs +1.68 EM gain	C1, C3, C4	undefined	The two baselines differ in whether gold context is present, so the two gains are not commensurable.
NOISE-05	rerun	AstuteRAG vs. RobustRAG	+4.4 accuracy in favor of ASTUTE RAG	C1, C5	survives	Model, dataset, retrieval and metric match, so the 4.4-point accuracy gap stands.
FAITH-02	quoted	FaithfulRAG vs. FaithEval (GPT-4o CF)	81.7% vs 47.5% (+34.2)	C1, C4	undefined	Aggregate and single-task scores under different conflict types and models cannot be differenced.
CALIB-03	invited	ADAPT-LLM vs. Self-RAG 7B	+16.75 accuracy in favor of Self-RAG	C1, C4	narrows	A harder test subset and a stricter metric are confounded with the method, so 16.75 is not a method effect.
Within-harness controls (one paper, one harness)						
FAITH-01	invited	RAGTruth (resp-level) vs. RAGTruth (span-level)	78.7% F1 vs span-level	C4	narrows	Response-level and span-level scores are different units, so the contrast holds only within one unit.
CALIB-01	stated	Self-RAG 7B vs. RAG-baseline	+23.9 accuracy in favor of Self-RAG	C3	survives	Within one paper and one harness the 23.9-point gap stands as reported.
CALIB-02	stated	CtrlA 7B vs. FLARE (reimpl.)	+4.0 accuracy in favor of CtrlA	C3	survives	Within one paper and one harness the 4.0-point gap stands as reported.

Table 5: Worked cross-paper comparisons. Of 12 comparisons published papers state (Src=stated) or invite by grouping (Src=invited), 7 have no common scope, 3 survive only on a strict subset and 2 survive as stated; 3 within-harness contrasts are negative controls. Every number is copied verbatim from a locally extracted full text and re-checked at its line by `verify_cross_paper_cases.py`. C1 evidence state, C2 origin/capability, C3 intervention, C4 outcome, C5 policy/utility. Appendix Table 6 gives source lines and coordinate-matched values.

ID	Side A (reported value)	Side B (reported value)	Coordinate-matched reading
SEC-01	PoisonedRAG black-box ASR 0.97 on NQ, GPT-4, 5 injected texts per target question, Contriever retriever, top-5 context	Corpus Poisoning Attack ASR 0.01 on NQ as re-run by PoisonedRAG (Table 4)…	not_recoverable_from_text
SEC-02	PoisonedRAG black-box ASR 0.97 on NQ, 5 injected texts per target question, GPT-4, Contriever	Phantom attack success 93.3% for Refusal-to-Answer on netflix trigger, Llama3-8B, Contriever, 1 poisoned document	PoisonedRAG ASR at 1 malicious text in top-5 = 0.48 on NQ black-box (Table 17 L2853-2856); Phantom = 93.3% at 1 document
SEC-03	RobustRAG: empirical ASR <=15% in all cells of Table II; vanilla-RAG ASR for the poisoning attack 37.0-86.0 and for prompt injection 66.0-100…	PoisonedRAG black-box ASR 0.97 on NQ, 5 malicious texts injected per target question into a 2,681,468-text database, GPT-4, Contriever	Within RobustRAG's own harness the poisoning reduction is 86.0 → <=15.0 at its worst cell and 37.0 → <=12.0 on NQ (Table II, certifiedRAG2024.txt L1480-1485 and L1502-1507)…
SEC-04	BadRAG MCOP 98.2% attack success with 10 adversarial passages; GPT-4 reject ratio 74.6% on NQ (denial-of-service)	PoisonedRAG black-box ASR 0.97 on NQ, GPT-4, 5 injected texts per target question	not_recoverable_from_text
SEC-05	AIP ASR 95.23% on MedSquad, 5 adversarial documents, GPT-3.5-Turbo for rephrasing	PoisonedRAG ASR 28.57% on MedSquad (as reported in AIP Table 1)	Both at 5 injected docs on MedSquad: AIP 95.23% vs PoisonedRAG 28.57%
NOISE-01	Self-CRAG accuracy 61.8% on PopQA, SelfRAG-LLaMA2-7b backbone	Self-RAG 7B accuracy 54.9% on PopQA	Self-CRAG 61.8 vs Self-RAG 54.9 on PopQA (same backbone, retriever, metric)
NOISE-02	CtrlA7B accuracy 61.8% on PopQA, Mistral-7B, BM25+BGE retriever	Self-RAG 7B accuracy 54.9% on PopQA, LLaMA2-7B, Contriever-MS MARCO	not_recoverable_from_text
NOISE-03	ChatGPT accuracy 76.00% at 0.8 noise ratio on English QA subset	ChatGPT3.5 average EM 71.47 under mixed noise on RAG-Bench	not_recoverable_from_text
NOISE-04	Chain-of-Note EM 41.83 vs Retrieve-Read EM 34.28 on NQ at 100% noise ratio, LLaMA-2 7B, DPR	RAAT EM 87.15 vs RALMmultiple EM 85.47 average across noise types, LLaMA2-7B, DPR	not_recoverable_from_text
NOISE-05	ASTUTE RAG accuracy 52.2% on NQ, Claude 3.5 Sonnet, real-world retrieval	RobustRAG keyword aggregation accuracy 47.8% on NQ, Claude 3.5 Sonnet (AstuteRAG Table 1)	ASTUTE RAG 52.2 vs RobustRAG 47.8 on NQ (matched LLM, dataset, metric)
FAITH-01	Finetuned Llama-2-13B response-level hallucination detection overall F1 78.7%	Same models at span-level (Table 6) — different and lower numbers for word-level localization	not_recoverable_from_text
FAITH-02	FaithfulRAG (mistral-7b-instruct) accuracy 81.7% on FaithEval; Self-RAG (Llama2-7B) 37.4% on FaithEval	GPT-4o 47.5% accuracy on counterfactual context task of FaithEval; GPT-4o 96.3% on closed-book QA	not_recoverable_from_text
CALIB-01	Self-RAG 7B accuracy 66.4% on TriviaQA-unfiltered, Contriever-MS MARCO	Llama2-7B with retrieval accuracy 42.5% on TriviaQA-unfiltered	Self-RAG 66.4 vs Llama2-7B RAG 42.5 on TriviaQA (all coordinates matched)
CALIB-02	CtrlA7B accuracy 76.4% on TriviaQA, Mistral-7B, BM25+BGE	FLARE accuracy 72.4% on TriviaQA, reimplemented with Mistral-7B	CtrlA 76.4 vs FLARE 72.4 on TriviaQA (matched backbone, retriever, dataset)
CALIB-03	ADAPT-LLM (SQuAD) accuracy 38.15% on PopQA test set, Llama-2 7B, Contriever, exact match	Self-RAG 7B accuracy 54.9% on PopQA long-tail subset (1,399 queries), Llama2-7B, Contriever-MS MARCO + web search	not_recoverable_from_text

Table 6: Reported values and coordinate-matched readings for every comparability case. The released ledger records the file and line range supporting each value; the verifier re-opens each range and fails on any mismatch.

D Census, Tier Control and Codebook Ablation

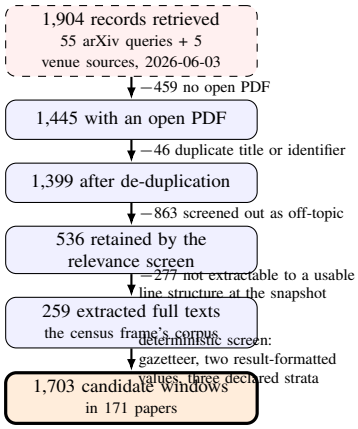


Figure 1: Sampling frame for the census. Solid boxes are stages a third party can re-execute from the release; the dashed box is documented but not re-executable, because per-query hit counts were not logged at the snapshot date and live indices drift. The frame is therefore reconstructible from the 259 released texts downward, and the population those texts were drawn from is not. Every composition figure in this paper is stated for the frame or for the re-weighted window universe, never for the literature.

Judgment	Ref. est.	Scored	Acc.	95% CI	Base	Acc. (LOO)
Provenance (transported / re-implemented / unclear)	171	170	0.91	[0.86, 0.96]	0.79	0.89
Verdict (survives / narrows / undefined)	129	114	0.82	[0.74, 0.89]	0.77	0.81
Per-class recall of the deployed panel, provenance						
transported		35	0.80	[0.64, 0.90]		
re-implemented		135	0.93	[0.88, 0.96]		
unclear		0	–			

Table 7: The deployed panel scored on *real* frame windows against a privileged-information reference, whose adjudicators read both papers' full texts rather than a 45-line window and whose every counted vote carries a quote located mechanically in the document that adjudicator was shown. "Ref. est." counts windows with a quote-verified majority; those without one are reported, not dropped. "Base" is the majority-class rate. The last column removes the one adjudicator sharing a checkpoint with the deployed panel. Measured sensitivity 1.00 and specificity 0.99 for the transported class make the composition correction non-trivial, unlike the identity map the submitted version used: it moves the re-implemented share to 0.881. This is not human gold.

Did the panel execute its own rule?	n	κ	95% CI
Verdict as *stated* by the judges	704	0.06	[-0.01, 0.11]
Verdict *composed* from their own coordinates	704	0.09	[0.01, 0.15]
Δ (composed − stated)		+0.02	[0.00, 0.05]
*Rule fidelity, judge by judge: share of codings where stated = composed, and count of **survives** verdicts the judge's own coordinates contradict*			
coderA_gpt5mini		1.000	0
coderB_claudeopus46		0.770	162
coderC_gemini31pro		0.925	53
What the evidence quotes are: 1,879 non-empty quotes classified			
verbatim in the passage shown		1,639	0.872
verbatim after Unicode/de-hyphenation		80	0.043
abridged with an ellipsis, fragments verbatim		76	0.040
table re-serialised from the passage		59	0.031
elsewhere in the same paper		–	0.000
re-serialised from elsewhere		4	0.002
near paraphrase of the passage		–	0.000
absent from the source document		21	0.011

Table 8: Two diagnostics the submitted version did not run, both over data it had already released. *Top:* the decision rule of §3 composed mechanically from each judge's own coordinate labels, beside the verdict that judge stated. Composing the rule does not make the panel reproducible, so the census failure is not the panel failing to execute its instrument. *Bottom:* every evidence quote classified against the passage its author was shown. The submitted version reported a single located rate, which pools an extraction artefact with a fabrication; separating them shows 0.987 of non-empty quotes are supported by the passage shown and 0.011 are absent from the source document altogether.

Field	Maj. share	Obs.	κ	$\kappa_{\max}$	$\kappa/\kappa_{\max}$
C1 evidence state	0.630	0.60	+0.19	0.72	0.26
C2 origin / capability	0.589	0.54	+0.11	0.61	0.18
C3 intervention	0.871	0.75	-0.11	0.42	-0.26
C4 outcome / unit	0.814	0.71	+0.08	0.53	0.16
C5 policy / utility	0.526	0.54	+0.10	0.51	0.19
Verdict	0.507	0.45	+0.06	0.46	0.13
The headline contrast, on matched label spaces					
Census, three classes		0.45	0.06	0.46	0.13
Census, binary		0.52	0.05	0.37	0.12
Constructed (stated), three classes		0.83	0.74	0.81	0.92
Constructed (stated), binary		0.95	0.86	0.88	0.97

Table 9: How much room κ had. $\kappa_{\max}$ is Cohen's maximum attainable κ, averaged over judge pairs: the largest value these judges could have reached without changing how often each of them uses each label (Umesh et al., 1989; Sim and Wright, 2005). A near-constant field caps κ regardless of behaviour, which is the argument §3 makes for C3; the last column applies it to every field instead of one, so the argument is symmetric. It also repairs the headline contrast: matching the label space does not close the gap between the constructed and the published setting, and the right reading of the census verdict is that its judges reach 0.13 of what their own marginals allow where the constructed judges reach 0.92.

Passage discloses …	Windows	Papers	Obs.	κ	95% CI
0 of four disclosed	44	9	0.66	0.10	[-0.14, 0.33]
1 of four disclosed	23	9	0.42	-0.11	[-0.32, 0.09]
2 of four disclosed	157	25	0.44	0.04	[-0.13, 0.09]
3 of four disclosed	253	60	0.42	-0.02	[-0.13, 0.05]
4 of four disclosed	227	57	0.44	0.08	[-0.07, 0.18]
High ($\geq$ 3)	480			0.03	
Low ($\leq$ 2)	224			0.11	
Difference (high − low)				-0.08	[-0.27, 0.05]

Table 10: The disclosure counterfactual on published text. Disclosure is scored deterministically from a released lexicon over the passage the judges were shown, so the exposure is a document feature and not a model output: regressing agreement on it is not regressing the panel's labels on themselves. Windows whose passages mention more coordinate-relevant terms do *not* yield more reproducible verdicts. This is the observational test of the checklist recommendation, and it does not support it; §6 shows what does, and what does not follow from it.

Panel	J	Common items		All	Own
		AC_1	κ	κ	κ
Submitted panel (1 mini, 2 front.)	3	0.29	0.14	0.33	0.07
Small tier only, 3 developers	3	0.16	0.00	0.29	0.00
Frontier only, 3 developers	3	0.54	0.31	0.51	0.25
Frontier only, 3 dev. $\times$ 2 checkpoints	6	0.47	0.22	0.38	0.23
same dev. (Anthropic)	2	0.56	0.35	0.51	0.49
same dev. (Google)	2	0.54	0.29	0.38	0.24
same dev. (OpenAI)	2	0.53	0.20	0.50	0.21
across dev. (pair A)	2	0.56	0.32	0.53	0.37
across dev. (pair B)	2	0.37	0.07	0.15	0.24

Table 11: Upgrading the judges helps and does not rescue the verdict, on a population that is identical for every row. The same 1,703-window frame is coded by a fully crossed pool of nine judges—a small and two frontier checkpoints from each of three developers—from which these panels are drawn. Because κ is prevalence sensitive, comparing panels each scored on its own majority-eligible subset would confound capability with item composition, so *Common items* fixes one population of 843 windows that a majority of the whole nine-judge pool called eligible, and *All* drops nothing at all: it scores every one of the 1,680 commonly coded windows, carrying ineligibility as its own category. *Own* is each panel's κ recomputed on the majority-eligible subset of *that panel*, which is the population a reader who ran only that panel would see; it is reported because it is what a smaller pool can obtain, and it is the column that must *not* be compared across rows. Conclusions are unchanged across the three: an all-frontier panel of three developers multiplies verdict κ several-fold over the submitted panel and still reaches only *fair*. The three *same dev.* rows are the reference level, two frontier checkpoints from one developer where nothing differs but the checkpoint; they bound what *correlated* judges reach, not what the task affords, and Table 12 shows that one cross-developer pair beats two of them.

Frontier pair	Dev.	κ	AC_1	Elig. κ
opus-4.6 vs opus-4.8	same	0.35	0.56	0.56
opus-4.6 vs gem-3.1p	cross	0.32	0.56	0.61
gem-3.1p vs gpt-5.4	cross	0.32	0.53	0.67
gem-3.1p vs gem-2.5p	same	0.29	0.54	0.50
opus-4.6 vs gpt-5.4	cross	0.29	0.52	0.61
opus-4.8 vs gpt-5.4	cross	0.25	0.50	0.43
gem-2.5p vs gpt-5.4	cross	0.22	0.47	0.52
opus-4.6 vs gem-2.5p	cross	0.20	0.47	0.33
gpt-5.4 vs gpt-5.5	same	0.20	0.53	0.71
gem-3.1p vs opus-4.8	cross	0.19	0.44	0.43
opus-4.8 vs gpt-5.5	cross	0.11	0.50	0.26
opus-4.8 vs gem-2.5p	cross	0.07	0.37	0.16
gem-2.5p vs gpt-5.5	cross	0.07	0.42	0.64
gem-3.1p vs gpt-5.5	cross	0.03	0.34	0.63
opus-4.6 vs gpt-5.5	cross	0.00	0.35	0.49

Table 12: Every frontier-frontier judge pair, on the common item population, ordered by verdict κ. The averaged same-developer advantage is +0.11; its exact randomisation over the 15 ways of matching six checkpoints into three disjoint pairs gives $p = 0.07$, against a smallest attainable value of 0.07. The contrast is also not uniform: 6 cross-developer pairs agree better than the weakest same-developer pair, so sharing a developer is reported as residual structure and not as an effect.

Instrument	AC_1	κ	Unan.	*Survives* share	$\Delta\kappa$
Five coordinates (released)	0.54	0.31	0.48	0.40–0.63	—
C1 and C4 only	0.56	0.34	0.51	0.55–0.62	+0.03 [-0.05, +0.12]
No coordinates, asked directly	0.58	0.31	0.53	0.62–0.73	-0.00 [-0.10, +0.11]

Table 13: The verdict does not become reliable when the instrument is simplified. The same three frontier judges, one per developer, code the same 1,703-window frame three times, changing only the codebook: the released five coordinates, the two coordinates that are not degenerate across systems, and no coordinates at all. All three arms are scored on the same 832 windows. $\Delta\kappa$ is against the five-coordinate arm with a 95% interval resampling source papers. *Survives* share is the range across the three judges of how often each calls a comparison sound; the spread, not the noise, is what fails to port.

Subset	N	Accuracy	95% CI
Panel majority, all items	238	0.945	[0.92, 0.97]
passage declares provenance	79	1.000	[1.00, 1.00]
provenance only from structure	80	1.000	[1.00, 1.00]
recall, reimplemented	80	1.000	[1.00, 1.00]
recall, transported	79	1.000	[1.00, 1.00]
recall, unclear	79	0.835	[0.75, 0.91]
Constant baseline	238	0.336	—

Table 14: The provenance label scored against a reference, not against agreement. Items are constructed excerpts whose provenance is fixed when the item is written; the prompt, the three judges and the majority rule are byte-identical to the ones that produced the census labels. The three reference classes are equinumerous, so the constant baseline is $1/3$. The *implicit* rows carry no declaration sentence at all, so a classifier that keyed on a re-implementation or attribution phrase would score at baseline there. The stratum that asserts a relation while supplying no number for the other side is not scorable under a three-option prompt and is reported separately: the deployed prompt sends 100% of those judgments to *unclear*, while the same prompt with a fourth option recovers them at 100%, so the category is codable and the deployed instrument simply cannot express it.

Quantity	Estimate	95% CI
Windows in the universe	6,710	—
surfaced by the screen	1,703	—
missed, sampled	420	—
Eligible rate, missed windows	0.336	[0.27, 0.41]
Eligible rate, screened windows	0.444	—
Screen recall, transported	0.498	[0.26, 1.00]
Screen recall, re-implemented	0.276	[0.24, 0.33]
Re-implemented share, in frame	0.874	—
Re-implemented share, re-weighted	0.926	[0.86, 0.96]

Table 15: What the screen misses. The universe is every window under the screen's own windowing that names another paper's system and carries at least one result-formatted number; the screen's second-numeral requirement and its three surface strata are dropped, which are exactly the conditions that select for baseline tables. A uniform sample of the missed windows was coded by the same three judges under the same two prompts. Intervals resample source papers.

E Constructed Instrument Benchmark

Item construction. Every specification is instantiated twice, once with one dedicated sentence per coordinate in the codebook's vocabulary and once in paper-style prose, so presentation is a paired within-specification contrast; the two instances state exactly the same facts. Table 16 in §6 reports every cell of the resulting design.

Window	12 lines of `poisonedRAG2024.txt`: a baseline table putting 0.97 beside 0.01 for the Corpus Poisoning Attack.
Screen	Gazetteer system absent from this paper's own title region; two result-formatted values in-window; *baseline* stratum. Enters the frame. No judgment yet.
Provenance	*re-implemented*: the 0.01 is this paper's measurement of the other system, not a value copied from it.
Coordinates	C1 **match** (same corpus, retriever, questions); C2 **match** (five injected texts on both sides); C3 **mismatch**, as always—*excluded from the rule*; C4 **mismatch** (target questions answered with the attacker's answer, against queries with an adversarial passage in the top-k); C5 **not determinable** (neither side states an action set or a cost).
Rule	C4 unmatched and no operating point at which the two quantities coincide ⇒ **undefined**. Replace with the common scope both texts support; do *not* substitute a computed number.
A panel	Nine judges agree on the provenance row (κ 0.55) and not on the rule row (κ 0.06). On constructed prose that *states* C5 they agree at κ 0.74: the disagreement lives in the coordinate that reads *not determinable*.

Figure 2: One juxtaposition traced end to end. The screen is deterministic; provenance is the judgment automation recovers; the rule row is the one it does not. The C5 label is where the failure of §6 lives: faced with a silent passage, judges answer something rather than *not determinable*.

Prose	Passage	N	Base	Coord.	Rule	Verd.	κ
Cano.	States every coordinate	90	0.33	0.97	0.891	0.822	0.73
Cano.	Withholds one, verdict unaffected	87	0.34	0.89	0.889	0.847	0.83
Cano.	Withholds one, verdict forced	36	1.00	0.85	0.403	0.380	0.26
Natu.	States every coordinate	90	0.33	0.93	0.835	0.791	0.75
Natu.	Withholds one, verdict unaffected	90	0.33	0.85	0.832	0.837	0.85
Natu.	Withholds one, verdict forced	35	1.00	0.81	0.252	0.252	0.21

Table 16: Disclosure crossed with prose realism, each cell scored against *its own* constant baseline (*Base*). *Cano.* states one coordinate per sentence in the codebook's own vocabulary; *Natu.* states the identical facts in paper-style prose, reordered and interleaved with sentences that carry no coordinate information. *Coord.* is mean recovery over C1, C2, C4 and C5. In the second row of each block the withheld coordinate does not determine the verdict, so all three classes occur and the baseline is $1/3$; in the third the verdict follows from the withholding alone, so the baseline is 1.00 and no estimator can beat it.

Estimator	Exact	Range	Weakest judge
Five coordinates (rule)	0.861	0.80–0.92	+0.46
Rule with C2/C5 ablated	0.738	0.70–0.78	+0.37
Two coordinates (rule, C1/C4 prompt)	0.832	0.80–0.87	+0.47
Judge verdict, five-coordinate prompt	0.824	0.76–0.89	+0.43
Judge verdict, C1/C4 prompt	0.832	0.80–0.87	+0.46
Judge verdict, no coordinates	0.796	0.74–0.86	+0.40
Majority class (narrows)	0.336	—	—

Table 17: Every estimator arm, on the 357 non-degenerate items, scored against the constant baseline of that population (0.336; the three verdict classes are equinumerous inside every disclosure arm, so this baseline is not an artefact of pooling). *Range* is across judges and *weakest judge* is the smallest judge margin over that baseline. The degenerate withholding arm, whose reference verdict never varies and whose constant baseline is therefore 1.00, is excluded here and reported separately in Table 16.

F Formal Estimand Definitions

Let reviewer r receive prompt $T_r \in \{0, 1\}$, rate item i, and emit label $Y_{ri}(T_r)$. Let Z_i denote a constructed reference and $s(Y, Z)$ a reference-relative score.

Realized-panel contrast.

$$\Delta_{\text{panel}} = \frac{1}{I} \sum_i \big[s(M_{r \in \mathcal{R}_1} Y_{ri}(1), Z_i) - s(M_{r \in \mathcal{R}_0} Y_{ri}(0), Z_i) \big],$$

where M is a majority rule and $\mathcal{R}_t$ is the realized reviewer panel. This quantity is descriptive for those panels and that reference.

Reviewer-population contrast.

$$\Delta_{\text{rater}} = \mathbb{E}_{r,i,Z}[s(Y_{ri}(1), Z_i) - s(Y_{ri}(0), Z_i)] .$$

This target requires treatment assignment and uncertainty at the reviewer level, plus a defined reviewer population and reference distribution.

Consequence contrast.

$$\Delta_{\text{loss}} = \mathbb{E}[L\{a(Y(1)), \theta\} - L\{a(Y(0)), \theta\}] ,$$

where $a(\cdot)$ is the accept, narrow, reject, or defer decision and θ is the deployment state.

Variance decomposition. Let $\mathcal{R}$ be the realized reviewer composition and $\mathcal{Z}$ the realized reference process. Then

$$\operatorname{Var}(\widehat{\Delta}) = \mathbb{E}\Big[\operatorname{Var}(\widehat{\Delta} \mid \mathcal{R}, \mathcal{Z})\Big] + \operatorname{Var}\Big[\mathbb{E}(\widehat{\Delta} \mid \mathcal{R}, \mathcal{Z})\Big] .$$

Source resampling with reviewers and references fixed estimates the first term. Reviewer assignment and reference construction contribute to the second.

Assignment asymptotics for panel majorities. For three conditionally independent votes with correctness probabilities p_1, p_2, p_3, majority correctness has expectation

$$m(p_1, p_2, p_3) = p_1p_2 + p_1p_3 + p_2p_3 - 2p_1p_2p_3.$$

Let Z be a six-versus-six reviewer assignment and $\mu(Z) = \mathbb{E}(\widehat{\Delta} \mid Z)$ the expected difference between the two scheduled panel-majority scores. With I conditionally independent items (or G independent source clusters of bounded size),

$$\operatorname{Var}(\widehat{\Delta}) = O(I^{-1}) + \operatorname{Var}_Z\{\mu(Z)\}.$$

The source-only standard error targets the first term. If reviewer profiles are heterogeneous, the nonlinear function m generally makes $\mu(Z)$ nonconstant, so the second term does not vanish when more items are added to the same twelve assignments. Conditional item precision can therefore converge to zero while uncertainty over reviewer assignment remains. This characterization explains the direction of the simulation; it does not supply a population model for reviewer abilities.

G Assignment-Unit Simulation Details

The simulation is a two-arm review study with 12 reviewers, six per prompt, G sources contributing six claims each, and a balanced three-of-six panel per claim in both arms; baseline agreement

Setting	Panel/src	Votes/src	Rev. t	Rev. RI
Size: no prompt effect, $\alpha = .05$				
$\sigma_r = 0.00, G = 80$	4.7	5.1	4.7	4.3
$\sigma_r = 0.27, G = 80$ *	24.2	32.9	4.6	4.3
$\sigma_r = 0.78, G = 80$	62.6	70.1	5.1	5.0
$\sigma_r = 0.27, G = 160$	36.7	46.5	4.4	4.3
Power: true prompt effect 0.4 *logits*				
$\sigma_r = 0.27, G = 80$	78.8	89.0	52.8	51.6
$\sigma_r = 0.78, G = 80$	69.6	77.0	11.4	11.4

Table 18: Rejection percentages over 5,000 replicates with twelve reviewers, six per prompt, six claims per source G, and balanced three-of-six panels. Reviewer abilities are redrawn every replicate, so the target is a reviewer-population effect rather than a realized-panel contrast. σ_r is the between-reviewer ability standard deviation on the logit scale; the starred row is the value estimated from the crossed three-coder diagnostic ($\hat{\sigma}_r = 0.27$, Appendix Table 21). *Panel/src* is the status-quo paired source t test on panel-majority correctness; *Votes/src* keeps the source-level standard error but uses individual votes; *Rev. t* is a two-sample t test on the twelve individual reviewer accuracies; *Rev. RI* refers the same contrast to all 924 arm assignments. The last two are executable retrospectively from released pre-aggregation votes with reviewer identifiers, and retain power.

is 75% and source and claim logit standard deviations are .15 and .35. Reviewer abilities are redrawn from $N(0, \sigma_r^2)$ in every replicate, so the target is a reviewer-population effect and a source-only analysis cannot be defended as answering a realized-panel estimand instead; each cell uses 5,000 replicates at $\alpha = .05$. *Panel/src* is the status quo, a paired source t test on panel-majority correctness computable from released majority labels alone; *Votes/src* keeps the source-level standard error but uses individual votes; *Rev. t* tests the twelve reviewer accuracies and *Rev. RI* refers it to all $\binom{12}{6} = 924$ arm assignments. The last two need only pre-aggregation votes with reviewer identifiers. Adding sources makes the status quo worse at every positive σ_r, because item precision improves while the number of prompt assignments stays at twelve, and replacing majority labels with individual votes does not help: the defect is the unit over which uncertainty is computed, not the aggregation. Both reviewer-level analyses hold the nominal level across the grid, with size between 3.7% and 5.7%, and their power is the binding constraint rather than their calibration.

Pool	Kind	$\hat{\sigma}_r$	95% CI	κ	$\kappa_{\max}$
API-served only	API	0.37		0.05	0.42
Open-weight only	open	0.51		0.09	0.46
Pooled	both	0.50	[0.33, 0.62]	0.10	0.48
Per-judge accuracy on the constructed benchmark, which is what σ_r is computed from					
insJ_gpt55	API	0.889			
apk_gem35flash	API	0.874			
insC_gem31pro	API	0.858			
apk_opus47	API	0.838			
insH_gem25pro	API	0.828			
insI_gpt54	API	0.828			
apk_sonnet46	API	0.813			
apk_gpt5mini	API	0.801			
insB_opus46	API	0.786			
insG_opus48	API	0.764			
apk_haiku45	API	0.749			
apk_gpt4o	API	0.722			
apk_gpt41	API	0.709			
qwen25-72b	open	0.814			
qwen3-32b	open	0.800			
qwen3-14b	open	0.794			
phi4	open	0.786			
qwen3-8b	open	0.728			
mistral-small-24b	open	0.719			
glm4-9b	open	0.603			
olmo2-13b	open	0.559			
internlm3-8b	open	0.554			

Table 19: Widening the reviewer pool. The submitted version estimated σ_r from six API-served judges, giving an interval spanning a factor of four. Adding 9 open-weight checkpoints, run under greedy decoding in one harness on byte-identical prompts, brings the pool to 22 and the interval to a factor of 1.9. It also shows the narrow pool was biased downward rather than merely imprecise: the API judges occupy a narrow accuracy band, and checkpoints reaching lower accuracies raise the estimate. The last two columns show the census result is not a property of one model class: the open pool fails to agree on the verdict as thoroughly as the API pool, and not because its labels are too uniform for κ to find signal.

Crossed design	Rev.	R	Items	$\hat{\sigma}_r$	Boot.
Ours: census judges (this task)	model	9	840	0.38	[0.18, 0.51]
Ours: constructed benchmark	model	6	432	0.38	[0.12, 0.48]
Ours: model panel	model	28	600	0.54	[0.31, 0.75]
DynaSent (sentiment)	human	674	122,117	0.78	[0.74, 0.81]
GoEmotions (neutral)	human	76	56,958	0.46	[0.37, 0.54]
LeWiDi (offensiveness)	human	116	8,377	0.45	[0.37, 0.53]
MultiPref (quality)	human	160	6,502	0.31	[0.24, 0.37]
Wiki Detox (toxicity)	human	3,944	159,686	0.55	[0.53, 0.56]

Table 20: Between-reviewer heterogeneity estimated with one moment estimator on every crossed reviewer-by-item design we could obtain. R is the number of reviewers retained (≥ 30 judgments for the public corpora). The first two rows are the estimate this paper needs and the literature does not supply: between-reviewer heterogeneity on cross-paper comparability adjudication itself, once on the real corpus against a leave-one-out majority of the other judges and once on the constructed benchmark against a reference fixed before any reviewer saw the item. Public human corpora are scored against a leave-one-out majority of the other annotators; the model panel is scored against the deterministic minimum of a displayed loss rule, which no reviewer supplies. The last column is the 2.5–97.5 percentile interval of a reviewer bootstrap (resampling reviewers with replacement, $B \geq 2{,}000$ draws), which targets the correct unit of replication for a between-reviewer variance parameter. An item bootstrap and a parametric bootstrap are reported in the supplement JSON for comparison.

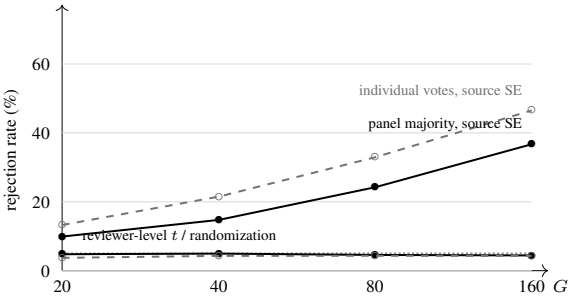


Figure 3: Null rejection rate against the number of sources G at the estimated reviewer heterogeneity $\sigma_r = 0.27$, with reviewer abilities redrawn each replicate. Adding sources makes the two source-level analyses steadily worse, and using individual votes instead of majority labels does not help: the defect is the unit of uncertainty, not the aggregation. The two reviewer-level analyses stay at the nominal level and are executable from released pre-aggregation votes. Table 18 gives the values and the corresponding power.

Quantity	Value
Crossed design	3 coders $\times$ 80 claims
Coder accuracy vs. adjudicated	0.850, 0.775, 0.725
$\hat{\sigma}_r$ (moment estimate)	0.271
Reviewer bootstrap 95%	[0.000, 0.347]
Item bootstrap 95%	[0.000, 0.522]
Parametric bootstrap 95%	[0.000, 0.732]
Parametric bias of $\hat{\sigma}_r$	-0.044
Identification from published reporting	
Widest admissible σ_r set	100% of the grid
Point identified by reporting?	no

Table 21: Reviewer heterogeneity is estimable only from a crossed reviewer-by-item design, and three coders do not estimate it. All three resampling schemes are shown because they disagree about the width and not about the conclusion: every interval contains zero, so this design alone cannot distinguish a badly miscalibrated panel from a homogeneous one. The reviewer bootstrap is the scheme reported for all seven designs in Table 20, because the reviewer is the unit the parameter is about; the item bootstrap resamples the wrong unit and is shown only for comparison. A parametric bootstrap puts the estimator's bias at -0.044 here, against at most 0.012 in every large design, which is what two degrees of freedom costs. The published agreement rates located in the audited papers place no usable restriction on σ_r: at the label prevalence those papers imply, every value on a 0–1.2 grid reproduces the reported agreement for some item dispersion. Separating the two components requires the item-by-annotator assignment, which none of the nine audited papers reports.

Abilities	Quantity	σ_r	G	Claims	Panel/src	Votes/src	Rev. t	Rev. RI
fixed_quantiles	size	0.00	20	120	4.9	5.2	4.9	3.7
fixed_quantiles	power	0.00	20	120	34.7	55.4	50.2	45.0
fixed_quantiles	size	0.00	40	240	4.7	4.9	4.9	4.0
fixed_quantiles	power	0.00	40	240	61.8	85.6	79.1	76.2
fixed_quantiles	size	0.00	80	480	4.9	4.9	4.5	4.0
fixed_quantiles	power	0.00	80	480	90.0	99.2	97.3	96.8
fixed_quantiles	size	0.00	160	960	4.9	4.6	4.2	3.8
fixed_quantiles	power	0.00	160	960	99.6	100.0	100.0	100.0
fixed_quantiles	size	0.23	20	120	8.5	10.9	5.0	3.9
fixed_quantiles	power	0.23	20	120	35.5	54.0	35.0	30.9
fixed_quantiles	size	0.23	40	240	13.3	18.3	4.9	4.4
fixed_quantiles	power	0.23	40	240	59.5	77.8	47.9	45.3
fixed_quantiles	size	0.23	80	480	20.6	28.7	4.7	4.2
fixed_quantiles	power	0.23	80	480	80.5	90.7	58.1	56.3
fixed_quantiles	size	0.23	160	960	31.9	42.2	5.2	4.7
fixed_quantiles	power	0.23	160	960	92.2	96.8	66.1	65.1
fixed_quantiles	size	0.27	20	120	9.7	13.2	4.8	3.9
fixed_quantiles	power	0.27	20	120	36.9	53.9	31.5	28.4
fixed_quantiles	size	0.27	40	240	16.1	22.5	4.8	4.2
fixed_quantiles	power	0.27	40	240	58.9	75.1	42.0	39.7
fixed_quantiles	size	0.27	80	480	25.9	34.5	4.9	4.6
fixed_quantiles	power	0.27	80	480	78.2	88.0	48.7	47.2
fixed_quantiles	size	0.27	160	960	39.1	48.4	5.2	4.8
fixed_quantiles	power	0.27	160	960	90.3	95.1	54.8	54.0
fixed_quantiles	size	0.31	20	120	12.4	15.7	4.9	3.9
fixed_quantiles	power	0.31	20	120	38.7	54.0	28.2	25.5
fixed_quantiles	size	0.31	40	240	18.9	25.6	4.9	4.5
fixed_quantiles	power	0.31	40	240	58.2	73.0	34.5	32.3
fixed_quantiles	size	0.31	80	480	31.3	41.0	5.3	5.0
fixed_quantiles	power	0.31	80	480	76.1	86.5	39.8	38.4
fixed_quantiles	size	0.31	160	960	43.9	53.4	5.1	4.7
fixed_quantiles	power	0.31	160	960	87.5	92.1	43.3	42.2
fixed_quantiles	size	0.40	20	120	15.9	21.3	5.0	4.3
fixed_quantiles	power	0.40	20	120	38.1	52.6	21.5	19.1
fixed_quantiles	size	0.40	40	240	26.3	36.0	5.2	4.8
fixed_quantiles	power	0.40	40	240	56.8	70.1	25.5	24.0
fixed_quantiles	size	0.40	80	480	39.8	49.6	4.6	4.1
fixed_quantiles	power	0.40	80	480	73.9	82.1	27.9	26.6
fixed_quantiles	size	0.40	160	960	53.5	62.9	5.3	4.9
fixed_quantiles	power	0.40	160	960	81.9	87.3	27.4	26.8
fixed_quantiles	size	0.45	20	120	18.7	26.1	5.1	4.2
fixed_quantiles	power	0.45	20	120	39.2	52.6	19.3	17.6
fixed_quantiles	size	0.45	40	240	29.6	39.1	4.8	4.3
fixed_quantiles	power	0.45	40	240	58.2	69.7	21.9	20.4
fixed_quantiles	size	0.45	80	480	45.0	53.9	4.9	4.5
fixed_quantiles	power	0.45	80	480	71.1	78.3	22.3	21.2
fixed_quantiles	size	0.45	160	960	57.8	66.5	5.1	4.9
fixed_quantiles	power	0.45	160	960	81.1	85.9	23.5	22.9

Table 22: Complete grid for the retrospective-remedy simulation (1 of 4). *fixed_quantiles* holds reviewer abilities at twelve population quantiles as in the original simulation; *resampled* redraws them each replicate. Entries are rejection percentages. The grid is split across floats because a single one exceeded a page and was previously clipped; the machine-readable grid is `supplement/retrospective_remedy_simulation.json`.

Abilities	Quantity	σ_r	G	Claims	Panel/src	Votes/src	Rev. t	Rev. RI
fixed_quantiles	size	0.46	20	120	19.2	26.2	4.7	4.2
fixed_quantiles	power	0.46	20	120	39.3	52.5	18.0	16.4
fixed_quantiles	size	0.46	40	240	31.0	40.9	4.8	4.5
fixed_quantiles	power	0.46	40	240	57.0	68.4	20.8	19.6
fixed_quantiles	size	0.46	80	480	46.6	56.0	5.5	5.0
fixed_quantiles	power	0.46	80	480	71.3	79.2	22.3	21.5
fixed_quantiles	size	0.46	160	960	59.0	66.9	5.2	4.9
fixed_quantiles	power	0.46	160	960	81.0	85.7	22.1	21.2
fixed_quantiles	size	0.53	20	120	23.9	32.2	5.0	4.3
fixed_quantiles	power	0.53	20	120	40.5	53.2	14.5	13.2
fixed_quantiles	size	0.53	40	240	36.7	47.0	5.4	4.9
fixed_quantiles	power	0.53	40	240	56.7	67.0	16.7	15.9
fixed_quantiles	size	0.53	80	480	49.5	58.7	4.8	4.5
fixed_quantiles	power	0.53	80	480	70.0	77.7	16.8	16.1
fixed_quantiles	size	0.53	160	960	64.0	71.7	5.5	5.3
fixed_quantiles	power	0.53	160	960	78.6	83.8	17.7	17.1
fixed_quantiles	size	0.54	20	120	23.6	31.4	5.2	4.5
fixed_quantiles	power	0.54	20	120	41.8	53.2	14.5	13.3
fixed_quantiles	size	0.54	40	240	36.6	46.2	5.0	4.8
fixed_quantiles	power	0.54	40	240	55.6	66.3	15.9	15.0
fixed_quantiles	size	0.54	80	480	51.1	60.9	5.1	4.8
fixed_quantiles	power	0.54	80	480	71.1	78.3	16.6	15.8
fixed_quantiles	size	0.54	160	960	63.3	70.7	4.9	4.7
fixed_quantiles	power	0.54	160	960	78.7	83.3	16.7	16.1
fixed_quantiles	size	0.55	20	120	24.5	31.9	4.6	4.0
fixed_quantiles	power	0.55	20	120	39.3	52.8	15.2	14.1
fixed_quantiles	size	0.55	40	240	37.9	48.4	4.6	4.3
fixed_quantiles	power	0.55	40	240	57.4	67.7	16.7	16.1
fixed_quantiles	size	0.55	80	480	51.0	60.5	4.3	4.1
fixed_quantiles	power	0.55	80	480	69.7	77.1	15.7	15.1
fixed_quantiles	size	0.55	160	960	64.8	72.0	5.4	5.3
fixed_quantiles	power	0.55	160	960	78.3	83.5	16.0	15.7
fixed_quantiles	size	0.78	20	120	35.5	44.4	5.2	4.4
fixed_quantiles	power	0.78	20	120	46.0	56.3	9.9	9.1
fixed_quantiles	size	0.78	40	240	52.2	60.7	4.9	4.7
fixed_quantiles	power	0.78	40	240	60.5	69.1	10.6	10.0
fixed_quantiles	size	0.78	80	480	64.1	70.7	5.1	4.9
fixed_quantiles	power	0.78	80	480	71.8	79.0	11.1	10.8
fixed_quantiles	size	0.78	160	960	75.1	80.3	5.6	5.5
fixed_quantiles	power	0.78	160	960	78.6	83.6	10.0	9.6
resampled	size	0.00	20	120	5.6	5.3	5.2	4.0
resampled	power	0.00	20	120	33.9	53.9	49.4	44.6
resampled	size	0.00	40	240	5.0	5.0	4.8	4.0
resampled	power	0.00	40	240	62.9	86.3	79.7	77.2
resampled	size	0.00	80	480	4.7	5.1	4.7	4.3
resampled	power	0.00	80	480	89.9	99.2	97.5	97.0
resampled	size	0.00	160	960	5.9	5.2	4.6	4.1
resampled	power	0.00	160	960	99.5	100.0	100.0	100.0

Table 23: Complete grid for the retrospective-remedy simulation (2 of 4). *fixed_quantiles* holds reviewer abilities at twelve population quantiles as in the original simulation; *resampled* redraws them each replicate. Entries are rejection percentages. The grid is split across floats because a single one exceeded a page and was previously clipped; the machine-readable grid is `supplement/retrospective_remedy_simulation.json`.

Abilities	Quantity	σ_r	G	Claims	Panel/src	Votes/src	Rev. t	Rev. RI
resampled	size	0.23	20	120	9.3	11.7	5.1	4.1
resampled	power	0.23	20	120	35.1	52.9	34.6	31.5
resampled	size	0.23	40	240	12.8	16.6	4.5	4.0
resampled	power	0.23	40	240	60.5	77.9	49.8	47.3
resampled	size	0.23	80	480	20.6	27.9	4.9	4.7
resampled	power	0.23	80	480	82.0	91.5	60.4	58.8
resampled	size	0.23	160	960	32.2	41.3	5.0	5.0
resampled	power	0.23	160	960	92.6	96.3	67.8	66.8
resampled	size	0.27	20	120	9.9	13.3	4.8	3.7
resampled	power	0.27	20	120	36.8	53.9	32.1	28.6
resampled	size	0.27	40	240	14.7	21.5	5.0	4.3
resampled	power	0.27	40	240	58.9	75.8	42.7	41.0
resampled	size	0.27	80	480	24.2	32.9	4.6	4.3
resampled	power	0.27	80	480	78.8	89.0	52.8	51.6
resampled	size	0.27	160	960	36.7	46.5	4.4	4.3
resampled	power	0.27	160	960	89.8	94.9	56.3	55.4
resampled	size	0.31	20	120	11.7	15.6	4.8	4.2
resampled	power	0.31	20	120	38.4	54.4	30.2	27.2
resampled	size	0.31	40	240	18.8	26.6	4.8	4.3
resampled	power	0.31	40	240	58.5	74.3	37.3	35.4
resampled	size	0.31	80	480	30.2	38.3	4.9	4.7
resampled	power	0.31	80	480	76.9	86.9	44.3	43.1
resampled	size	0.31	160	960	41.5	51.8	4.7	4.6
resampled	power	0.31	160	960	88.2	92.5	46.2	46.1
resampled	size	0.40	20	120	15.3	21.8	4.9	4.2
resampled	power	0.40	20	120	38.1	54.4	23.3	21.5
resampled	size	0.40	40	240	26.4	34.5	4.8	4.4
resampled	power	0.40	40	240	58.0	70.7	26.5	25.3
resampled	size	0.40	80	480	38.6	49.0	5.7	5.4
resampled	power	0.40	80	480	73.2	81.9	30.9	30.2
resampled	size	0.40	160	960	51.4	60.4	4.3	4.3
resampled	power	0.40	160	960	83.3	88.3	33.1	33.0
resampled	size	0.45	20	120	17.6	24.6	4.5	3.7
resampled	power	0.45	20	120	39.6	52.9	20.1	18.4
resampled	size	0.45	40	240	29.4	38.2	4.3	4.0
resampled	power	0.45	40	240	58.5	70.0	24.0	23.2
resampled	size	0.45	80	480	42.5	52.0	5.1	4.8
resampled	power	0.45	80	480	72.2	79.8	26.0	25.1
resampled	size	0.45	160	960	56.6	65.4	4.6	4.6
resampled	power	0.45	160	960	81.8	87.2	27.4	27.0
resampled	size	0.46	20	120	18.6	25.3	4.6	4.1
resampled	power	0.46	20	120	38.8	54.1	19.3	17.5
resampled	size	0.46	40	240	30.0	39.8	4.8	4.4
resampled	power	0.46	40	240	57.5	67.7	23.4	22.4
resampled	size	0.46	80	480	43.6	53.3	4.7	4.4
resampled	power	0.46	80	480	71.9	79.2	24.7	24.3
resampled	size	0.46	160	960	57.2	65.3	5.0	4.9
resampled	power	0.46	160	960	81.9	86.8	25.7	25.5

Table 24: Complete grid for the retrospective-remedy simulation (3 of 4). *fixed_quantiles* holds reviewer abilities at twelve population quantiles as in the original simulation; *resampled* redraws them each replicate. Entries are rejection percentages. The grid is split across floats because a single one exceeded a page and was previously clipped; the machine-readable grid is `supplement/retrospective_remedy_simulation.json`.

Abilities	Quantity	σ_r	G	Claims	Panel/src	Votes/src	Rev. t	Rev. RI
resampled	size	0.53	20	120	23.3	30.8	5.3	4.5
resampled	power	0.53	20	120	41.4	54.3	16.7	15.3
resampled	size	0.53	40	240	35.8	45.7	4.7	4.4
resampled	power	0.53	40	240	58.0	68.8	20.0	19.1
resampled	size	0.53	80	480	48.8	57.3	5.1	5.1
resampled	power	0.53	80	480	70.2	78.5	19.9	19.6
resampled	size	0.53	160	960	63.2	70.8	4.7	4.5
resampled	power	0.53	160	960	79.7	85.1	20.5	20.5
resampled	size	0.54	20	120	23.2	31.4	5.0	4.5
resampled	power	0.54	20	120	41.1	53.4	16.6	15.5
resampled	size	0.54	40	240	36.6	46.7	5.4	5.2
resampled	power	0.54	40	240	56.7	67.7	18.8	18.2
resampled	size	0.54	80	480	49.6	58.3	5.1	4.9
resampled	power	0.54	80	480	70.6	78.5	19.3	18.7
resampled	size	0.54	160	960	63.0	70.6	4.9	4.8
resampled	power	0.54	160	960	79.2	84.9	19.7	19.6
resampled	size	0.55	20	120	23.7	32.9	4.8	4.2
resampled	power	0.55	20	120	41.4	53.5	16.5	15.0
resampled	size	0.55	40	240	36.3	45.9	4.8	4.4
resampled	power	0.55	40	240	56.9	67.9	17.7	17.4
resampled	size	0.55	80	480	50.8	58.9	4.8	4.5
resampled	power	0.55	80	480	70.7	78.2	19.6	19.2
resampled	size	0.55	160	960	62.7	70.8	4.6	4.6
resampled	power	0.55	160	960	79.6	84.2	18.7	18.9
resampled	size	0.78	20	120	35.0	44.0	4.8	4.3
resampled	power	0.78	20	120	46.6	56.6	10.7	9.9
resampled	size	0.78	40	240	49.9	58.4	5.1	5.0
resampled	power	0.78	40	240	59.5	68.1	11.8	11.8
resampled	size	0.78	80	480	62.6	70.1	5.1	5.0
resampled	power	0.78	80	480	69.6	77.0	11.4	11.4
resampled	size	0.78	160	960	73.0	79.2	5.6	5.6
resampled	power	0.78	160	960	80.0	85.5	11.9	12.3

Table 25: Complete grid for the retrospective-remedy simulation (4 of 4). *fixed_quantiles* holds reviewer abilities at twelve population quantiles as in the original simulation; *resampled* redraws them each replicate. Entries are rejection percentages. The grid is split across floats because a single one exceeded a page and was previously clipped; the machine-readable grid is `supplement/retrospective_remedy_simulation.json`.

Setting	Source	Panel RI	Rater RI	SD/SE
$\sigma = 0.20$, $G = 20$	7.6 (0.4)	4.9 (0.3)	3.7 (0.3)	1.14
Homog., $G = 80$	4.9 (0.3)	4.5 (0.3)	4.5 (0.3)	0.99
$\sigma = 0.20$, $G = 80$	17.0 (0.5)	5.3 (0.3)	4.6 (0.3)	1.44
$\sigma = 0.20$, $G = 160$	25.9 (0.6)	4.9 (0.3)	5.2 (0.3)	1.77

Table 26: Oracle design benchmark, retained for reference. Entries are rejection percentages with Monte Carlo standard errors in parentheses, and SD/SE is the empirical standard deviation of the D−C contrast divided by the mean source-only standard error, so a value above one marks an optimistic standard error. Each row uses 5,000 replicates, 12 auditors (six per prompt), six claims per source, and balanced three-of-six panels; G is the number of sources and rater ability follows 12 fixed normal quantiles on the logit scale around 75% mean agreement. Source t and Panel RI use the identical studentized panel-majority statistic; Panel RI refers it to all 924 six-versus-six assignments under a full-vote oracle with an arm-independent schedule, and Rater RI uses individual accuracy. Panel RI is not implementable retrospectively, which is why Table 18 replaces it with two analyses that are, and evaluates them at the estimated σ_r rather than at an assumed one.

Rater profiles	Sources	Claims	Source-only rejection	Panel-RI rejection	Rater-RI rejection	SD/SE
$\sigma_r = 0.20$	20	120	7.6%	4.9%	3.7%	1.14
$\sigma_r = 0.20$	40	240	10.3%	4.5%	4.3%	1.22
Homogeneous	80	480	4.9%	4.5%	4.5%	0.99
$\sigma_r = 0.10$	80	480	8.0%	5.3%	4.5%	1.13
$\sigma_r = 0.20$	80	480	17.0%	5.3%	4.6%	1.44
$\sigma_r = 0.30$	80	480	28.3%	4.8%	4.8%	1.84
$\sigma_r = 0.20$	160	960	25.9%	4.9%	5.2%	1.77

Table 27: Complete null-simulation sweep. Each row uses 5,000 replicates and a true zero prompt effect. Panel RI uses the same studentized panel-majority statistic as the source-only test. A raw-contrast randomization sensitivity and Monte Carlo standard errors are recorded in the released JSON. The four headline rows appear in Table 26.

H Crossed Deployment-Panel Details

Developer (models)	Probe	C2 Δ	C5 Δ
01.AI (1)	+12.3	-5.2	+2.5
AI2 (2)	+2.3	-4.3	-1.6
Alibaba (5)	-0.2	+10.4	+2.1
Anthropic (2)	+0.4	-2.3	-7.0
Google (2)	+0.9	+6.6	-3.2
HuggingFace (2)	-0.1	+2.7	-3.7
IBM (2)	-3.9	-1.3	+0.1
Meta (1)	-4.8	+5.6	-3.5
Microsoft (2)	-0.4	+10.2	+3.1
Mistral (2)	-7.0	-0.6	+0.0
NVIDIA (1)	+1.0	+4.7	-2.0
TII (2)	-1.8	+1.7	+5.0
Zhipu (1)	-0.8	-4.7	+5.4
Pooled (random effects)	-0.4	+3.1	+0.1
95% CI	[-1.2, +0.4]	[+0.7, +5.5]	[-1.3, +1.5]
95% pred.	[-4.4, +3.6]	[-9.8, +16.0]	[-7.2, +7.3]
I^2	90%	98%	97%
dev. level	+0.1	+1.7	+0.4

Table 28: Controlled consequence test, *full* minus *placebo* in accuracy points, aggregated by developer. Checkpoints are nested in developers, so the developer-level row averages within developer before pooling. The prediction interval, not the confidence interval, is the quantity relevant to a claim about a model population; it contains zero on both substantive strata. The competence probe is the negative control. Per-model contrasts, Holm-adjusted values, paraphrase spread and the decision-normalisation report are in Appendix Table 29.

Model	Stratum	Abl.	Plac.	Full	F−P	F−A	Para. SD	McNemar p	Holm p
Qwen2.5-14B	all	60.9	59.0	61.6	+2.6	+0.7	3.3	0.008	–
Qwen2.5-14B	probe competence	94.8	91.9	92.3	+0.4	-2.5	5.5	0.031	–
Qwen2.5-14B	c2 capability	59.9	56.6	61.4	+4.8	+1.5	5.8	0.031	0.000
Qwen2.5-14B	c5 utility	44.9	44.9	46.5	+1.6	+1.6	2.4	0.001	0.068
Qwen2.5-7B	all	60.0	56.0	59.8	+3.7	-0.2	2.4	0.003	–
Qwen2.5-7B	probe competence	89.2	85.4	82.1	-3.3	-7.1	8.8	1.000	–
Qwen2.5-7B	c2 capability	66.1	59.0	70.1	+11.1	+4.0	7.0	0.000	0.000
Qwen2.5-7B	c5 utility	39.3	38.3	38.2	-0.1	-1.0	2.1	0.383	1.000
Qwen3-14B	all	54.1	54.8	61.3	+6.5	+7.2	4.3	0.000	–
Qwen3-14B	probe competence	94.4	96.5	97.3	+0.8	+2.9	2.8	1.000	–
Qwen3-14B	c2 capability	53.5	51.7	65.9	+14.3	+12.4	5.1	0.000	0.000
Qwen3-14B	c5 utility	34.5	37.2	38.8	+1.6	+4.3	5.1	0.002	0.253
Qwen3-8B	all	59.7	60.0	65.9	+5.9	+6.2	4.2	0.000	–
Qwen3-8B	probe competence	69.0	69.2	70.0	+0.8	+1.0	4.0	1.000	–
Qwen3-8B	c2 capability	74.0	74.5	80.5	+6.0	+6.6	7.9	0.016	0.000
Qwen3-8B	c5 utility	40.8	40.9	49.2	+8.2	+8.3	8.6	0.000	0.000
OLMo-2-13B	all	47.4	46.0	47.8	+1.8	+0.4	5.8	0.016	–
OLMo-2-13B	probe competence	69.6	65.0	70.6	+5.6	+1.0	10.5	0.016	–
OLMo-2-13B	c2 capability	50.7	48.3	53.1	+4.8	+2.4	3.1	0.000	0.000
OLMo-2-13B	c5 utility	32.9	34.2	31.0	-3.1	-1.9	9.2	0.021	0.000
OLMo-2-7B	all	41.5	40.2	41.7	+1.4	+0.1	7.8	0.500	–
OLMo-2-7B	probe competence	43.1	48.1	44.8	-3.3	+1.7	4.5	0.500	–
OLMo-2-7B	c2 capability	44.8	46.9	44.3	-2.6	-0.5	5.5	1.000	–
OLMo-2-7B	c5 utility	37.5	29.7	37.5	+7.8	+0.0	15.6	1.000	–
Granite-3.1-8B	all	53.8	54.8	53.7	-1.2	-0.1	2.3	0.011	–
Granite-3.1-8B	probe competence	81.9	85.2	78.1	-7.1	-3.7	7.7	0.003	–
Granite-3.1-8B	c2 capability	55.9	56.4	57.6	+1.2	+1.7	3.5	1.000	0.200
Granite-3.1-8B	c5 utility	37.5	38.1	37.5	-0.6	+0.0	0.7	1.000	0.213
Phi-4-mini	all	61.5	60.8	62.0	+1.2	+0.5	0.7	0.000	–
Phi-4-mini	probe competence	67.3	69.4	69.8	+0.4	+2.5	7.5	1.000	–
Phi-4-mini	c2 capability	75.8	73.2	74.0	+0.7	-1.9	4.7	0.024	1.000
Phi-4-mini	c5 utility	44.2	44.0	46.0	+2.1	+1.9	2.1	0.000	0.213
Falcon3-7B	all	58.4	58.1	59.1	+1.0	+0.7	2.1	0.860	–
Falcon3-7B	probe competence	66.7	66.7	66.7	+0.0	+0.0	0.0	1.000	–
Falcon3-7B	c2 capability	68.8	68.3	69.0	+0.6	+0.2	1.2	1.000	0.901
Falcon3-7B	c5 utility	44.0	43.5	45.5	+2.0	+1.6	4.3	0.678	0.302

Table 29: Per-model, per-stratum results for the version-2 controlled experiment. Accuracies are percentages averaged over four paraphrases; F−P and F−A are the full-minus-placebo and full-minus-ablated contrasts in percentage points; *Para. SD* is the standard deviation of the full-minus-placebo contrast across paraphrases. McNemar tests use per-item paraphrase-majority decisions. Holm adjustment covers gate-passing models on the two substantive strata.

I Framework and Taxonomy Details

Adjacent line	Primary unit	Unresolved comparison question	This paper's increment
General RAG surveys (Gao et al., 2024; Sharma, 2025)	Architectures, modules, applications	Whether similarly named gains concern the same evidence state, outcome, and experimental game	Adds a claim-level comparison lens and an estimand audit; does not claim a venue-complete census.
Evaluation surveys and unified comparisons (Yu et al., 2024; Wang et al., 2024; Alushi et al., 2026)	Metrics, benchmarks, and common-harness reruns	Which controls are required by a specific cross-study or mechanistic inference when rerunning is impossible	Separates scientific coordinates from assignment, participant, item, and reference units; the simulation quantifies one resulting failure mode.
Trustworthy and security RAG surveys (Zhou et al., 2024; Ni et al., 2025; Xu et al., 2026)	Trust dimensions, attack surfaces, and defenses	Whether benign conflict and attacker-created corruption share a transferable capability envelope	Makes origin/access/budget explicit and shows why a prompt-overlapping reference cannot establish independent criterion validity.
Conflict and abstention surveys (Xu et al., 2024; Wen et al., 2025)	Conflict types and refusal behavior	Whether a confidence or sufficiency signal identifies the action that maximizes utility at a matched operating point	Separates signal, action, and utility and releases a consequence-defined counterfactual target without claiming an evaluated policy.

Table 30: Positioning relative to adjacent surveys and empirical comparison work. Each adjacent line partitions the literature by mechanism, which is what a reader needs to *find* related work. None partitions the *comparison*, which is what a reader needs when two numbers appear together, and none is accompanied by evidence that its categories can be read off a paper. The increment is a reporting-coordinate reduction that is tested (§6) and applied at scale (§5); it is not a venue-complete systematic review, and it does not supersede these surveys for discovery.

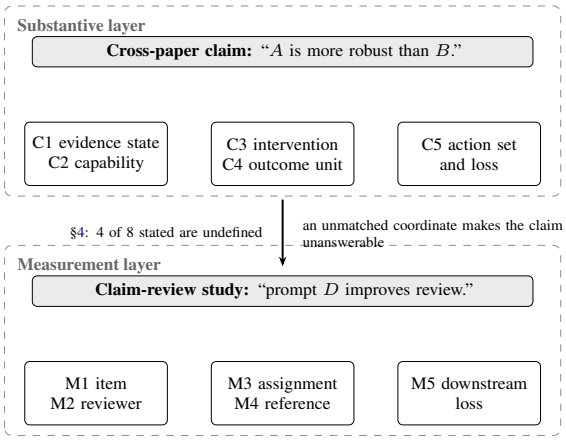


Figure 4: The two layers answer different questions about the same sentence. The substantive layer asks whether two published results concern one decision problem; Table 5 applies it to real comparisons. The measurement layer asks what a study of claim review estimates; Table 18 shows which analyses of such a study are valid. Passing one layer cannot repair failure in the other.

Field	Question	Representative values	Why it changes a comparison
Evidence state	What is observable at the decision boundary?	Supported; distracting; insufficient; conflicting; stale/weak provenance; adversarially manipulated; not evidence-conditioned.	Query ambiguity and model uncertainty are not silently mixed with properties of retrieved evidence.
Origin / capability	How did that state arise, and what can an attacker observe or change?	Benign retrieval; constructed stressor; natural drift/conflict; distribution shift; attacker-created evidence with access, budget, and target.	Transfer depends on the data-generating process and the attacker's capability envelope.
Intervention	What part of the system changes?	Corpus/index; retriever; reranker; context assembly; generator; verifier; system control; evaluation artifact.	Components may be alternatives in a shared harness or complementary stages in a pipeline.
Outcome	What object and unit are measured?	Answer correctness; retrieval quality; claim support; attack success; privacy leakage; calibration; selective risk; cost.	Identically named datasets do not make different target quantities interchangeable.
Policy / utility	What response is selected, under which costs and stakes?	Answer; verify; re-retrieve; expose conflict; clarify; quarantine; recover; abstain; coverage–risk utility.	A signal becomes operational only through an action set and a declared utility context.
Control check	Does the experiment support the intended inference?	Dataset and corpus snapshot; retriever; top-k; generator; decoding; budget; baselines; uncertainty.	Tuple compatibility alone cannot remove experimental confounding.

Table 31: The diagnostic schema. The five fields identify whether two results address a compatible question; a claim-specific control check then determines whether their reported numbers support the intended inference. The fields are not assumed orthogonal or sufficient.

Coordinate	Controlled values	Equivalence / boundary rule	Verdict effect
Evidence state	Supported; irrelevant/distracting; insufficient; conflicting; stale/weak provenance; adversarially manipulated; not evidence-conditioned.	Code what is observable at the decision boundary. Query ambiguity, OOD input, confidence, and attacker access are not evidence states by themselves.	A direct performance claim normally requires a common state or an explicit cross-state estimand.
Origin / capability	Benign retrieval; constructed stressor; natural conflict/drift; query or distribution shift; attacker-created evidence; shared clean harness.	Record how the state arose. For security, include corpus write access, query control, retriever visibility, generator access, insertion budget, objective, and target.	A cross-source rank over two different games narrows at best and is otherwise undefined; a shared harness may deliberately vary capability as the controlled factor.
Intervention point	Corpus/index; retriever; reranker; context assembly/compression; generator prompting or training; verifier; system-control policy; evaluation-only artifact.	Code the component changed. Different interventions may be compared as alternatives when a shared end-to-end harness controls the remaining pipeline; otherwise they provide component-specific or compositional evidence.	C3 never enters the decision rule. Comparing two different systems means the intervention differs, so a C3 mismatch is the normal state of a comparison and cannot by itself make one undefined; it is recorded so a reader knows what changed.
Outcome / unit	Answer correctness; retrieval quality; evidence support; attack or defense utility; privacy leakage; calibration; selective risk/coverage; cost/latency; human preference.	Record definition, direction, aggregation, and unit. Relative gains over different baselines are not automatically a shared estimand.	A one-scalar ranking requires a common target. Without one there is no operating point to re-read at, so the gap is undefined rather than narrowed.
Policy / utility	Answer; abstain; re-retrieve; filter; verify; expose conflict; clarify; quarantine; recover; not applicable; unreported.	Use *not applicable* for fixed descriptive comparisons. When policies compete, report action set, stakes, preferences, coverage, and cost.	Different actions can be ranked only under common utility; a signal alone does not establish the preferred response. A stated price at the other side's setting narrows the comparison.
Second-stage controls	Dataset/corpus snapshot; retriever/top-k/chunking; generator/decoding; compute budget; baselines; uncertainty/statistical design.	Select controls according to the stated inference. The tuple cannot remove confounding from unmatched experimental systems.	Tuple compatibility is necessary diagnostic information but not sufficient for a direct empirical ranking.
Final verdict	Survives; narrows; undefined.	Exactly the rule of §3, and the only verdict vocabulary used anywhere in this paper or its artifacts. *Survives*: C1, C2, C4 and C5 all match, so the quoted gap stands. *Narrows*: a threatening coordinate differs but a side reports an operating point at the other's setting, or the comparison holds on a strict subset; report that matched number. *Undefined*: no such operating point exists, or a decision-critical field is unavailable after full-text inspection.	Verdicts attach to comparison claims, not permanently to paper pairs, and a new shared-harness experiment can change the supported verdict. An undefined comparison is never replaced by a substitute number computed across the unmatched coordinate.

Table 32: The decision procedure of §3 in full. The five coordinates diagnose semantic compatibility and claim-specific controls determine whether the claim as written is warranted, but only C1, C2, C4 and C5 enter the verdict: C3 is the thing being compared and is recorded rather than tested. The three verdicts here are the same three used in the body, the census codebook and every released artifact.

Boundary case	Five-coordinate diagnosis	Second-stage control check	Supported conclusion
Different interventions, shared harness	A retriever filter and a generator-side reader address the same evidence state, metric target, and answer policy; intervention differs.	Same corpus, retriever output before intervention, generator, budget, metric, and seeds.	A valid end-to-end comparison is possible. Intervention mismatch alone does not force *partial*.
All five fields match, systems differ	Two answer-only methods target benign retrieval noise and answer correctness at the generator.	Different datasets, retrievers, model scales, top-k, and compute budgets.	The papers are conceptually comparable but their reported numbers are not directly rankable without a shared rerun or adjustment.
Action unnamed but observable	A QA benchmark never uses "action policy" but requires one answer for every item and gives no abstention channel.	Evaluation code and output schema confirm answer-only behavior.	Code *answer* when policy is decision-relevant; use *not applicable* for a fixed descriptive comparison.
Same pair, different claim	An attack artifact and a defense share a threat model but optimize attack success and defended utility.	A common attack–defense harness can evaluate both roles; a paper-only scalar comparison cannot.	Valid as components of one evaluation, partial as substitutable methods, invalid if ranked under incompatible objectives.
Same label, incompatible capability	Two poisoning studies both report attack success, but one assumes bounded corpus insertion and the other controls retrieved instructions at query time.	Capability, budget, retrieval probability, and objective do not align.	A direct ranking is invalid even if intervention, metric name, and answer behavior appear similar.

Table 33: Counterexamples delimiting the schema. Verdicts depend on the stated comparison and experimental controls; the five coordinates are diagnostic fields rather than a necessary-and-sufficient classifier.

Threat class	Corpus	Query	Retr.	Gen.	Objective	Boundary
Dense-retriever poisoning (Zhong et al., 2023)	write	train set	white/gray	no	retrieve bad passages	Not always end-to-end.
RAG knowledge corruption (Zou et al., 2025; Xue et al., 2024)	write	target topic	black/gray	black	answer steering	Stealth/dynamics vary.
Backdoor-style RAG (Cheng et al., 2024; Chaudhari et al., 2024)	write	trigger	gray	black	latent behavior	Normal queries may pass.
Indirect prompt injection (Chaturvedi et al., 2025; Chang et al., 2026)	write/share	natural	black	black	instruction hijack	Must overcome retrieval.
Opinion/fairness manipulation (Chen et al., 2025b; Bagwe et al., 2025)	write	group/topic	black/gray	black	bias steering	QA metrics miss social harm.
Defense setting (Xiang et al., 2024; Yin et al., 2026; Choudhary et al., 2025; Zhang et al., 2025a)	bounded	varies	assumed	black/gray	detect/demote/trace	Depends on budget/visibility.

Table 34: Attacker-capability matrix for adversarial RAG. The same "attack success rate" can mean different things depending on corpus write access, query control, retriever visibility, generator access, and objective.

J Evaluation and Reporting

Required field	What to report	If omitted	Example consequence
Evidence-state definition	Supported, irrelevant/distracting, insufficient, conflicting, stale/weak provenance, adversarially manipulated, or not evidence-conditioned.	Papers with the same robustness label may test different evidence properties.	Noise training is ranked against sufficiency prediction.
Origin / capability	Benign retrieval, constructed stressor, natural conflict/drift, distribution shift, or attacker access, budget, objective, and target.	Stressors are not portable.	Natural conflict is compared with poisoning.
Intervention boundary	Corpus/KB, retriever/context, context assembly, generator, system control, or evaluation artifact.	Pipeline modules are treated as substitutable.	A benchmark is ranked against a mitigation method.
Retrieval setting	Corpus snapshot, retriever, top-k, chunking, reranking/compression, and index drift controls.	Results cannot be reproduced or normalized.	A filter looks better because its retriever retrieved easier negatives.
Outcome and unit	Answer correctness, support, attack success, privacy leakage, selective risk, calibration, cost, or action utility, including definition and aggregation.	Different objectives collapse into one score.	Lower attack success hides clean-utility loss.
Policy and utility	Answer, abstain, retrieve again, filter, verify, expose conflict, clarify, quarantine, or recover, plus stakes, preferences, coverage, and cost when actions compete.	Accuracy and confidence are over-interpreted.	A calibrated answerer may still answer from insufficient evidence.
Model and comparison controls	Dataset, generator, decoding, compute budget, baselines, seeds, uncertainty intervals, and significance procedure.	Tuple-aligned papers can still be empirically confounded.	A method appears better because it uses a stronger generator or a larger budget.
Uncertainty / abstention handling	Coverage, selective risk, confidence target, abstention reward, and re-retrieval cost.	Abstention can hide retrieval failure.	Refusal is rewarded when re-retrieval should be the correct action.
Security threat model	Attacker access, query control, retriever visibility, insertion budget, objective, and adaptive-defense assumptions.	Security numbers are not comparable.	Prompt injection defenses are claimed to protect against corpus poisoning.
Provenance of every quoted number	Whether the other side's figure was *transported* from its paper or *re-implemented* in this harness, and at which configuration.	The one property an automated audit recovers reliably (κ 0.55) is lost, and the stratum where incomparability lives cannot be found.	A re-implementation at a strengthened attack budget reads as a defeat of the published attack.
Reviewer/judge panel and unit map	Who or what adjudicated, how many, which items each saw, and how any prompt or instruction was assigned across them.	The unit of assignment is unrecoverable, so uncertainty cannot be computed at the unit that was varied.	A source-level test rejects a true null 38.6% of the time at the heterogeneity measured here.
Pre-aggregation votes	Per-reviewer labels with stable reviewer identifiers, before adjudication or majority voting.	Only reviewer-level analyses hold their nominal size, and they need these votes.	A reported agreement rate cannot be converted into a valid reviewer-level test after the fact.

Table 35: Reporting checklist for robust RAG under unreliable evidence, in the order of §7. The first block is the five coordinates and the controls that accompany them. §6 shows why they head the list: a comparison whose answer turns on a coordinate the paper does not state is the one case in which a panel neither agrees nor is right, and C1 and C4 are the two fields whose absence produces that case most often. The last block is what the audit of the audit requires.

Benchmark	Risk focus	Task / data	Perturbation or threat	Metrics	Boundary
RGB (Chen et al., 2024a)	noise, rejection, integration, counterfactuality	English/Chinese QA	noisy passages, false evidence, negative evidence	accuracy, ability-specific scores	Broad diagnostic benchmark but not adaptive attack-oriented.
RAGTruth (Niu et al., 2024)	hallucination / faithfulness	QA, data-to-text, summarization	naturally generated unsupported spans	word-level hallucination labels	Strong annotation granularity; less focused on retrieval-side threat models.
Sufficient Context (Joren et al., 2025)	answerability	QA with context sufficiency labels	insufficient retrieved evidence	selective generation / abstention	Captures answerability but not adversarial corruption.
MagicMushroom (Zhang et al., 2025c)	retrieval noise	configurable RAG stress tests	distracting, low-quality, inconsequential, irrelevant noise	robustness under noise types	Fine-grained noise; limited conflict/security coverage.
URAG (Nguyen et al., 2026)	uncertainty	RAG as MCQA / conformal prediction	irrelevant/noisy retrieval and uncertainty sets	coverage, set size, calibration	Good for uncertainty; less direct for free-form generation.
QueryRobust (Percin et al., 2025)	query robustness	QA with query perturbations	wording variation at query–retrieval interface	retrieval and end-task degradation	Focuses on query perturbation, not corpus corruption.
PoisonArena (Chen et al., 2025a)	adversarial poisoning	RAG attack competition setting	competing poisoning attacks	attack success, retrieval effects	Strong security focus; benign uncertainty is secondary.
SafeRAG (Liang et al., 2025)	RAG security	security benchmark for RAG	poisoning / malicious retrieval contexts	attack and defense metrics	Security-oriented; not a full evidence reliability suite.
MTRAG-UN (Rosenthal et al., 2026)	multi-turn unanswerability	multi-turn RAG conversations	unanswerable, underspecified, non-standalone questions	abstention / clarification quality	Covers interaction; less focused on adversarial evidence.
CRUMQs (Liu et al., 2025a)	unanswerable multi-hop RAG	realistic multi-hop queries	unanswerable and uncheatable questions	answerability and robustness	Useful for abstention; not a broad attack/noise benchmark.

Table 36: Benchmark-level comparison for robust RAG evaluation. The table makes explicit what each benchmark perturbs, what it measures, and which failure modes remain outside its scope.

Main claim	Cited evidence basis	What the evidence supports	Boundary
Irrelevant or noisy context can harm generation.	Irrelevant-context robustness, RAAT, RGB, and counterfactual-noise studies.	Noise affects both retrieval quality and generator context use.	Does not imply one robustness score covers all noise types.
Knowledge conflict differs from adversarial corruption.	Conflict and poisoning studies with different evidence origins and evaluation targets.	Natural disagreement and attacker-created contradiction require different assumptions.	Conflict-resolution results should not be read as poisoning-defense results.
RAG attacks require capability normalization.	Poisoning, backdoor, prompt-injection, traceback, and certification studies.	Attack/defense results require corpus access, query control, retriever visibility, and budget.	The paper does not rank attacks by real-world prevalence.
Calibration, abstention, and retrieval control are distinct.	Sufficient-context, uncertainty, adaptive-retrieval, and calibration studies.	Confidence estimation, evidence sufficiency, retrieval triggering, and abstention optimize different quantities.	Does not validate a single universal control policy.
Action-trace evaluation is a benchmark agenda.	Benchmarks exposing subsets of answer, evidence, action, cost, and threat fields.	Existing benchmarks cover pieces of the proposed schema.	No new leaderboard or empirical system ranking is claimed.

Table 37: Narrative claim-to-source ledger. Expanded and strict citation audits were model-assisted and deterministic; this table records the intended evidence boundary rather than certifying every citation.

K Scoping-Map Audit Artifacts

Evidence layer	Unit and size	Structured coverage	Supported use and provenance boundary
Scoping corpus	536 broadly included papers	Metadata and broad relevance	Documentary recall pool; single model-assisted broad screen, not duplicated and not prevalence-estimating.
Map B	342 paper rows	Literature family and intervention; free-text metrics; no structured evidence origin or policy	Source-stratified landscape synthesis; adjudicated subsets, but not a five-coordinate map.
Claim inventory	289 claims from purposive fixed sections	Model-coded coordinates, evidence flags, and claim-specific controls	Source-linked hypothesis index; its verdict distribution and internal ablations are not validated results.
Instrument benchmark	360 constructed comparability items	Five coordinate values and a verdict fixed when each item is written; six judges from three developers under three codebooks	The instrument claim of §6. Templated prose, so it bounds what the schema can do on text that states it, not on any particular published passage.
Cross-paper census	1,703 frame windows	Nine judges, fully crossed, one codebook	The prevalence and reliability claims of §5. No verdict-based rate is an estimate.

Table 38: The released evidence layers answer different questions. Larger model-assisted layers do not validate claim-level verdicts; the human layers test an annotation procedure rather than literature prevalence. No layer is treated as an independently validated gold standard.

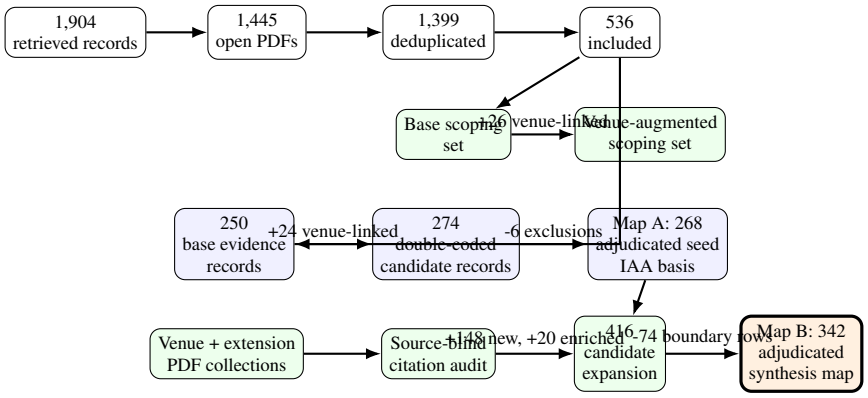


Figure 5: Corpus lineage. The 416-row venue-expanded candidate map is second-coded and adjudicated into Map B (342 rows), the canonical synthesis evidence map. Map A (268 rows) is the double-coded adjudicated seed used for IAA and adjudication audit. The base and venue-augmented scoping sets are used for boundary checks, while a separate 300-paper source-blind citation audit over the venue and extension PDF collections decides which papers enter the main text, case studies, or appendix.

Stage	Count	Rule
Initial retrieval	1,904 records	Keyword search over arXiv, ACL Anthology, OpenReview, USENIX, AAAI/ACM landing pages where open versions were available, plus curated robust/trustworthy RAG paper lists; search cutoff 2026-06-03.
Open full-text corpus	1,445 PDFs	Downloaded only open PDFs; removed failed downloads, inaccessible files, and non-PDF artifacts.
Deduplicated candidates	1,399 papers	Deduplicated by normalized title and arXiv/proceedings identifiers; retained the most complete metadata record when preprint and proceedings versions overlapped.
Relevance screening	536 included papers	Included papers that study RAG robustness, reliability, uncertainty, calibration, faithfulness, attack/defense, or evaluation under unreliable evidence; excluded application-only RAG papers.
Core pool	112 papers	Papers directly shaping the taxonomy, threat/failure model, intervention mechanisms, or evaluation framework.
Core reading set	100 papers	Core papers used as main narrative anchors; the remaining 12 core-pool papers stay in the broader scoping set.
Base scoping set	300 papers	Core papers plus representative supporting work across all taxonomy branches before venue follow-up triage.
Venue-augmented scoping set	326 papers	Base scoping set plus 26 venue-linked additions after follow-up triage.
Candidate evidence records	274 papers	Base evidence map of 250 papers plus 24 venue-linked additions, double-coded and adjudicated.
Map A adjudicated seed	268 papers	Candidate evidence records retained after final adjudication; used for IAA/adjudication provenance.
Map-B candidate expansion	416 papers	Map A plus 148 source-blind human-citation-audit additions and 20 enriched original rows; used for second-coding review.
Map B adjudicated synthesis map	342 papers	Map A plus 74 retained additions after second coding and final adjudication; frozen for current descriptive statistics. Post-freeze discoveries enter a recall log until a new version is coded.

Table 39: Screening-flow summary. Search ends on 2026-06-03; coding and adjudication artifacts freeze on 2026-06-15. The supplement provides query strings, source counts, rules, exclusions, hashes, and a post-freeze recall log.

Artifact / set	Count	Count definition	Use in this paper
Map-A double-coded candidates	274	Candidate records coded by two human sources before final inclusion adjudication.	Used for human-human agreement and adjudication; six records were excluded from Map A.
Map A adjudicated seed	268	Included records after adjudication: 42 venue/proceedings-linked, 222 arXiv-only or emerging preprints, 2 preprints with year tags, 1 review-platform record, and 1 other/uncategorized record.	Used as the IAA/adjudication provenance snapshot, not the current synthesis denominator.
Venue-augmented scoping set	326	Original base scoping set plus 26 triaged venue-linked additions.	Used for broader scoping and boundary checks; not all rows are evidence-map records.
Source-blind citation audit	300	Human citation decisions over venue PDFs and extension PDFs.	Determines which audited papers enter the main text, case studies, appendix, background, or exclusion set; distinct from the base scoping set.
Map-B candidate expansion	416	Map A plus 148 candidate additions supported by the source-blind citation audit; 20 original rows enriched with citation-audit metadata.	Pre-adjudication expansion candidate map used for second-coding review.
Map B adjudicated synthesis map	342	Map A plus 74 retained additions after second coding and final adjudication.	Current evidence map for source-stratified synthesis and Map-B cross-tabs.

Table 40: Reconciliation of reported corpus counts. The 342-row adjudicated Map B is the canonical synthesis map; Map A and the 274-candidate set are retained to report double-coding and adjudication provenance.